

ADAPTING PRIOR-DATA FITTED NETWORKS FOR TABULAR ANOMALY DETECTION

Maximilian Bershtman Niv Cohen

Faculty of Electrical and Computer Engineering
Technion – Israel Institute of Technology
Haifa, Israel

max.bershtman@gmail.com cohen.niv@gmail.com

ABSTRACT

While deep features have transformed anomaly detection in images and video, their impact on tabular data has been less substantial, partly due to the limited availability of strong deep representations. Recently, prior-data fitted networks (PFNs) have emerged as a promising source of such representations for tabular data. In this work, we investigate how PFN representations can be adapted and leveraged for anomaly detection.

The question is harder than it looks. No anomalies are available before deployment, so model parameters cannot be tuned with supervision, and the reference set that defines normal behavior may itself contain the very anomalies it is supposed to reveal. We begin our study using frozen TabPFN features. Scoring each sample by its distance to its nearest neighbors in feature space already gives strong results. We identify which layers to use and a feature-extraction procedure suited to the task. Next, to further improve performance, we use the reference set to fine-tune the model, so that the resulting features better separate normal samples from anomalies. On the ADBench benchmark, our fine-tuning free approach (ZEN) reaches a higher mean AUROC than every baseline, and our fine-tuned method (FOCUS) improves on it further. Our approach also generalizes across PFN models¹.

1 INTRODUCTION

Tabular data is the most common data format in real-world machine learning (Hollmann et al., 2023). This paper studies anomaly detection on tabular data, one of the oldest tasks in the field and still one of its most active (Han et al., 2022; Yoon et al., 2026; Li et al., 2026). The task is unsupervised: a detector receives a reference set of samples that describe normal behavior, and must score how strongly each new sample deviates from them. High scores should single out samples such as fraudulent transactions, unusual patient records, or compromised network traffic. The detector faces two unknowns at once. First, the anomalies: they are rare and their structure is unknown in advance, which is why the task cannot be posed as supervised classification. Second, the reference set: it may itself be contaminated, as when fraud investigators can verify only a small fraction of transactions and the unverified rest enter the historical record as normal (Dal Pozzolo et al., 2018).

We therefore study two settings. Our main setting is the *contaminated regime*, where the reference set contains anomalies at an unknown rate and nothing removes them before a detector sees it. The *clean regime*, where the reference set is anomaly-free but unlabeled, is the complementary setting.

Existing detectors handle these unknowns only partly. Detectors that excel on a clean reference set can degrade sharply once anomalies enter it (Marszałek et al., 2026), because an anomaly already in the reference set makes the test anomalies that resemble it look normal. Classical detectors remain remarkably strong (Liu et al., 2008; Zhao et al., 2019), but every deployment starts from zero, with nothing carried over. Deep methods learn richer representations (Livernoche et al., 2024; Yin et al., 2024; Li et al., 2025; Dai et al., 2025), but they too train one model per dataset, and their gains have

¹Code: github.com/Maximilianb1/zen-focus-tabular-anomaly-detection

not proven robust across datasets. The ADBench benchmark study reached a sobering conclusion: no unsupervised detector in its comparison came out reliably better than the others, “emphasizing the importance of algorithm selection” (Han et al., 2022).

Anomaly detection in images solved a similar problem through *transfer*: over the last decade it was rebuilt on representations reused from models pretrained once at scale (Yosinski et al., 2014; Bommasani et al., 2021). Features taken from generically pretrained networks and scored by simple k -nearest-neighbor distance outperformed elaborate self-supervised detectors (Bergman et al., 2020), and the same kind of features now underlies state-of-the-art industrial defect detection (Cohen & Hoshen, 2020; Roth et al., 2022). Tabular data could not follow this path, lacking comparably strong generic representations (Grinsztajn et al., 2022; Shwartz-Ziv & Armon, 2022). PANDA, an image anomaly detection method, names the absence of generic extractors for some data modalities as the main limitation of its approach (Reiss et al., 2021).

Recently, prior-data fitted networks (PFNs), with TabPFN chief among them, have emerged as strong pretrained models for tabular data, and as a potential candidate feature extractor. These models are transformers pretrained on synthetic tabular data (Hollmann et al., 2023; 2025; Grinsztajn et al., 2026), and they match or beat strong tuned baselines on supervised tabular prediction (Hollmann et al., 2025). They work by in-context learning: given a new table as context, they predict for query samples in a single forward pass, with no dataset-specific training.

Here, we ask how such a model can best be leveraged for tabular anomaly detection. So far, PFN-based work on the task has taken two routes. One pretrains a dedicated in-context detector on synthetic anomaly-detection tasks (Marszałek et al., 2026; Shen et al., 2025; Ding et al., 2026). The other keeps TabPFN frozen and uses it as a plug-in scorer over its predictions (Prior Labs, 2025; Ding et al., 2026). Adapting a pretrained PFN to the task itself remains mostly unexplored, although adaptation of PFNs has shown some success for supervised learning (Feuer et al., 2024). Notably, adaptation is especially interesting as a detector built by adapting TabPFN would inherit future improvements of the PFNs. Moreover, PFNs are never pretrained to find anomalies, so adapting one also measures how much its representations carry beyond their pretraining objective. This raises the question we study: can a general-purpose tabular foundation model be adapted into a competitive anomaly detector?

This paper shows the answer is yes. We reach it in two steps that adapt TabPFN, first without changing its weights and then by updating them (Figure 1). First, ZEN (*Zero-training Embedding Neighbors*) scores a test sample by its distance to its nearest neighbors in the frozen model’s embedding space. We develop a feature-extraction procedure suited to the task and weight each reference sample by how much it can be trusted, so the score stays reliable even when the reference set is contaminated. Second, FOCUS (*Fine-tuned One-Class Unsupervised Scoring*) fine-tunes the model so that its features better separate the normal data from anomalies. Our evaluation for the contaminated settings finds that ZEN already reaches a higher mean AUROC than every baseline in our comparison, and FOCUS the highest of all.

Contributions.

- We introduce ZEN, a training-free anomaly detector on frozen TabPFN embeddings whose score stays reliable even when the reference set is contaminated (Sec. 2.2).
- We propose FOCUS, which builds on ZEN: a trust-weighted compactness fine-tune of TabPFN that reaches the highest mean AUROC in both regimes (Secs. 2.3 and 3).
- Our approach is not tied to a single PFN model: applied without any change to four other tabular foundation models, it improves each of them significantly (Sec. 4).

2 METHOD

2.1 PRELIMINARIES

Prior-data fitted networks. A prior-data fitted network (PFN) is a transformer trained to perform supervised learning in a single forward pass (Hollmann et al., 2023). It receives a labeled dataset $D_{\text{train}} = \{(x_i, y_i)\}_{i=1}^n$ as its context and a query sample x_{test} . From these it outputs a predictive

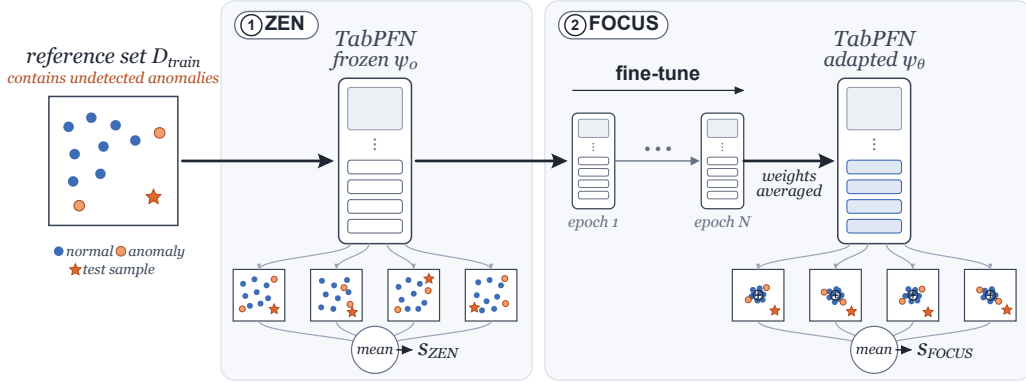


Figure 1: Overview of ZEN and FOCUS. The reference set contains undetected anomalies (orange). (1) ZEN embeds the reference set with frozen TabPFN, using only the last four of its 24 blocks (bottom slabs). Each test sample is scored by its distance to its nearest reference samples in each block’s embedding, and the four scores are averaged. (2) FOCUS fine-tunes the model so that the reference set’s embeddings contract toward the fixed center c , and averages the weights of its epoch checkpoints (two epochs) into one adapted model. The adapted embeddings are scored as in (1). The drawing is simplified; Secs. 2.2 and 2.3 give the full methods.

distribution $q_\theta(y \mid x_{\text{test}}, D_{\text{train}})$, taking no gradient step on D_{train} . In Bayesian terms, that forward pass approximates a posterior predictive distribution. A prior over data-generating mechanisms $\phi \in \Phi$ induces

$$p(y \mid x, D_{\text{train}}) \propto \int_{\Phi} p(y \mid x, \phi) p(D_{\text{train}} \mid \phi) p(\phi) d\phi. \quad (1)$$

While this integral is intractable, a PFN approximates it: synthetic datasets are sampled from the prior, and the network is trained to predict each dataset’s held-out labels from the rest of that dataset (Appendix D.1 states the objective). Pretraining happens once; every new table afterwards is handled in context. TabPFN uses a synthetic prior built on structural causal models, and it matches or beats strong tuned baselines (Hollmann et al., 2023; 2025).

TabPFN as a feature extractor. Beyond its predictions, TabPFN, like any transformer, computes an internal representation of every sample it processes, which we use in this work. Because it is an in-context model, every embedding depends on the whole reference set. In addition, TabPFN embeds a sample differently depending on whether it is provided as context or used as a query. We therefore embed each reference sample as a query against the rest of the set, leave-one-fold-out, following Ye et al. (2025). Appendix D.1 gives the details, and the labels used for TabPFN’s context.

Anomaly detection setting. The anomaly detection setting we study provides an unlabeled reference set $D_{\text{train}} = \{x_i\}_{i=1}^n \subset \mathbb{R}^d$, assumed mostly normal. It is evaluated on a test set D_{test} composed of normal samples and anomalies. A detector assigns each test sample a score $s(x)$, oriented such that higher means more anomalous. Ground-truth labels are used only for evaluation. The two regimes we investigate differ in one assumption about D_{train} : in the *clean regime* it contains no anomalies, and in the *contaminated regime* it contains anomalies at an unknown rate. Which regime holds is often unknown at deployment time.

2.2 ZEN: ZERO-TRAINING EMBEDDING NEIGHBORS

Anomalies tend to lie where normal data is sparse. The classical way to act on this is density-estimation based scoring: rate each test sample by how far it sits from the reference samples. The usual measure is the distance to the k nearest reference samples; it is one of the oldest anomaly detectors and, on clean tabular data, still one of the strongest (Sec. 3). Our approach ZEN applies a similar rule, but in the embedding space of the frozen TabPFN model. Embed every sample, then score each test sample by its mean distance to the k nearest reference samples:

$$s(x) = \frac{1}{k} \sum_{j \in N_k(x)} \|\psi(x_j) - \psi(x)\|_2, \quad (2)$$

where $\psi(x)$ is a sample’s embedding and $N_k(x)$ indexes the k reference samples whose embeddings are nearest to $\psi(x)$. We use $\psi_0^{(l)}(x \mid C)$ for the frozen TabPFN’s embedding of x after transformer block l , given the context C ; the index 0 stands for the original, frozen weights, as opposed to the fine-tuned weights θ of Sec. 2.3. TabPFN has 24 transformer blocks, and ZEN takes the score from the last four of them, a window we denote W (Appendix C.5 ablates this choice). I.e., Eq. 2 is computed on $\psi_0^{(l)}$ for each block $l \in W$. Distances in different blocks live on different scales, so each block’s scores are standardized before they are averaged. The mean and standard deviation come from the reference samples’ own leave-one-out distances. The average over the four blocks is the score of one run of the model. ZEN uses this score together with two techniques that handle a reference set containing anomalies, explained next.

PFN feature extraction. We can get a more informative score by embedding the data several times rather than once, each time with a different subset of the features. ZEN therefore runs the frozen model on $B+1$ *feature subsets*. B of them each hold a random half of the features, $S_1, \dots, S_B \subset \{1, \dots, d\}$, and the last one, S_{B+1} , uses all features together. Because the model uses its context to calculate embeddings, each feature subset yields a different representation of the data. To regulate the extracted features so that they sit on a similar scale, we also add to the context of each S_i synthetic samples drawn uniformly on $[-1, 1]$ (Appendix D.1 gives every constant of the extraction and Appendix C.8 ablates them).

Soft cleaning. Under contamination the reference set contains anomalies, and an anomalous test sample that lands next to one of them may receive a small k NN distance and pass as normal. We therefore weight the reference samples by how well their own neighbors support them. Let a_i be the mean distance of reference sample i to its k nearest other reference samples. Each of the $B+1$ feature subsets and four blocks yields one such distance. We standardize each over the reference set and average the values. The result is one isolation estimate $\hat{a}_i$ per reference sample, shared by every feature subset and block. Sample i receives the *trust weight* $w_i = \exp(-\lambda \hat{a}_i)$, and every distance to it is divided by w_i :

$$\tilde{s}(x) = \frac{1}{k} \sum_{j \in N_k(x)} \frac{\|\psi(x_j) - \psi(x)\|_2}{w_j}, \quad (3)$$

where $N_k(x)$ is the neighbor set of Eq. 2, found with the plain distance. A reference sample that sits far from its own neighbors, as an undetected anomaly typically does, receives a small weight: its distances are inflated, so a test sample next to it no longer looks normal, while the sample itself is never removed. At $\lambda=0$ every weight is one and the cleaning is off. Appendix C.6 provides a sensitivity study.

The scoring function. For each feature subset S_i , Eq. 3 is computed per block $l \in W$, standardized, and averaged over the four blocks, giving the score $s_i(x)$. ZEN averages the B random subsets and gives that average the same weight as the weight of the full set of features, S_{B+1} :

$$s_{\text{ZEN}}(x) = \frac{1}{B} \sum_{i=1}^B s_i(x) + s_{B+1}(x). \quad (4)$$

2.3 FOCUS: FINE-TUNED ONE-CLASS UNSUPERVISED SCORING

ZEN uses the embedding space exactly as pretraining left it. Yet, that space is shaped for supervised prediction on synthetic tasks, not for separating the normal samples of a given dataset from its anomalies. Image anomaly detection faced the same situation and moved past its frozen baselines by adapting pretrained features to the normal training data (Reiss et al., 2021; Reiss & Hoshen, 2023; Cohen & Avidan, 2022). FOCUS brings this adaptation to TabPFN. Following the fixed-center convention of DeepSVDD and PANDA (Ruff et al., 2018; Reiss et al., 2021), it fine-tunes the last blocks of the model such that the last layer embeddings of the reference set contract toward a fixed center c , the trust-weighted mean of the frozen model’s final-block embeddings of that set. Under contamination the reference set contains anomalies, so the trust weights enter the loss: each reference sample pulls toward the center in proportion to how much it can be trusted (according to the weights explained in *Soft cleaning* above). The weights and center are computed once from the

Table 1: Main results, sorted by contaminated-regime AUROC. Each regime has two columns: the mean AUROC ($\times 100$) over the 47 ADBench datasets, and the mean rank over the regime’s 18 methods. The best value per column is in bold and the second best is underlined. ^d marks a deep learning method, trained per dataset; ^t marks a TabPFN-based method.

Method	Contaminated		Clean		Method	Contaminated		Clean	
	AUROC	rank	AUROC	rank		AUROC	rank	AUROC	rank
FOCUS ^t (ours)	80.0	5.9	85.9	<u>4.9</u>	PCA	72.5	8.9	76.7	9.0
ZEN ^t (ours)	<u>79.7</u>	<u>6.0</u>	<u>85.2</u>	5.6	TCCM ^d	72.2	9.7	80.8	7.6
IForest	76.4	7.2	79.6	8.4	DeepSVDD ^d	70.9	11.4	74.7	11.8
TabPFN-PL ^t	74.6	8.8	78.8	9.4	OCSVM	70.1	11.4	75.6	10.8
COPOD	74.4	8.4	74.5	11.3	SOD	69.1	11.7	58.7	15.6
HBOS	74.4	8.6	77.4	10.0	COF	66.4	12.5	55.7	15.4
DTE ^d	74.3	9.8	81.5	7.9	LODA	66.0	12.7	69.8	12.9
raw k -NN	73.9	7.8	85.1	4.4	LOF	65.4	11.8	82.9	6.2
ECOD	73.9	9.1	74.2	12.0	CBLOF	73.7	9.2	81.7	7.8

frozen model and never updated. We write ψ_θ for the fine-tuned extractor. Appendix D.1 gives the loss and every detail of the fine-tune, and Appendix C.7 ablates its constants.

Averaging the fine-tuning epochs. Adapting pretrained features to the normal data can collapse them. DeepSVDD avoids the collapse by constraining the network and fixing the center (Ruff et al., 2018). PANDA reports that naive adaptation often deteriorates the features and counters it with early stopping combined with a regularizer (Reiss et al., 2021). Anomaly detection provides no labeled validation set that would allow us to decide how to regularize the compactness finetuning in a supervised manner. Here we take a simpler route: FOCUS trains for two epochs on every dataset and averages the weights saved after the first and after the second epoch, in the spirit of stochastic weight averaging (Izmailov et al., 2018). No further regularization is added. Once adapted, the fine-tuned model replaces the frozen one in every step of ZEN, using the same feature subsets, the same contexts, and the same soft-cleaned readout. We call the resulting method s_{FOCUS} .

3 RESULTS

3.1 EXPERIMENTAL SETUP

Data and protocol. We evaluate on all 47 classical tabular datasets of the ADBench benchmark (Han et al., 2022). They span 80 to 619,326 samples and 3 to 1,555 features (Appendix B.1). In the clean regime the training data (reference set) takes 70% of the normal samples and the test set holds the remaining normals and every anomaly, according to the one-class convention (Ruff et al., 2018; Li et al., 2025). In the contaminated regime a stratified 70/30 split preserves the anomaly rate on both sides, as in ADBench. Datasets above 10,000 samples are first subsampled to 10,000, stratified and without replacement. We also note a leak in the official pipeline, which duplicates the samples of the twelve datasets with fewer than 1,000 samples before the split, so that most test samples there have a twin in the training set. We fix that and evaluate these datasets at their natural size instead (Appendix A.2). Features are scaled to $[-1, 1]$ by a MinMax scaler fitted on the training split, identically for every method. Seeds 0 through 4 were fixed in advance, and each seed draws a fresh split. Appendix A gives the protocol in full.

Baselines. The comparison covers the 13 classical and deep detectors that ADBench itself benchmarks through PyOD (Han et al., 2022; Zhao et al., 2019). To these we add recent detectors with public code: TCCM (Li et al., 2025), DTE (Livernoche et al., 2024), the official unsupervised TabPFN extension (Prior Labs, 2025), and a TabPFN pseudo-labeling control (pseudo-labels are assigned using a PCA-based anomaly score). No method, ours included, is tuned per dataset: baselines run with their authors’ recommended settings, ours with one fixed configuration per regime. Every number is computed by us under this protocol; Appendix D gives every configuration.

Metrics. We report AUROC. AUPRC is reported in Appendix B.3 and wall-clock inference times in Appendix D.3. Per-dataset values are means over the five seeds. Appendix B.4 reports the spread. Each method is summarized by its mean AUROC and mean rank across the 47 datasets, and every two methods are compared with a Wilcoxon signed-rank test (Demšar, 2006).

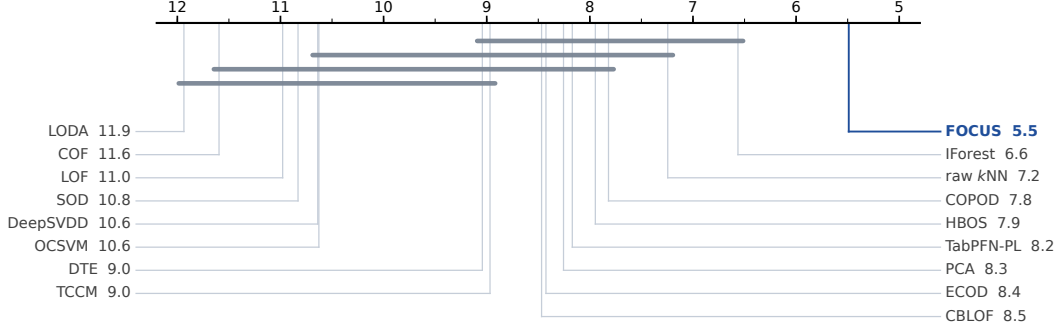


Figure 2: Contaminated regime: mean rank of FOCUS and of every ranked baseline over the 47 datasets, best rank on the right (we only report FOCUS here, so ranks differ from Table 1). FOCUS is tested against each of the 16 baselines with a Wilcoxon signed-rank test, Holm-corrected over the 16 comparisons. Bars among the baselines join methods whose pairwise differences are not significant. FOCUS has a statistically significant advantage in every comparison. Appendix C.1 details the construction.

3.2 MAIN RESULTS

Contaminated regime. Methods that perform very well in the clean regime often perform much worse under contamination. Table 1 shows the pattern in each method’s two regime columns. The nearest-neighbor detectors suffer the most: raw-feature k NN loses 11 AUROC points, and LOF loses 17. Isolation Forest degrades far less and becomes the strongest baseline. Our methods sit above it. ZEN, our training-free version, reaches a higher mean AUROC than every baseline, 3.3 points above Isolation Forest ($p = 0.02$). FOCUS improves on ZEN by a further 0.3 points and reaches the highest mean AUROC of all methods, 3.6 points above Isolation Forest ($p = 0.01$) and 6.1 above raw-feature k NN ($p = 0.02$). ZEN and FOCUS also hold the two best mean ranks. We compare FOCUS with every ranked baseline at once. For each comparison we use the Wilcoxon signed-rank test with a Holm correction, and find FOCUS significantly ahead of all the baselines (Figure 2). Against the deep and TabPFN-based detectors, the closest to ours in spirit, FOCUS leads by a wide margin, and ZEN’s margin over every baseline is also positive and significant (Figure 3). Under AUPRC (Table 8) FOCUS keeps the best mean in both regimes; Appendix C.2 adds per-regime critical-difference diagrams and the complete pairwise matrix.

Clean regime. When the reference set contains no anomalies, the picture changes: raw-feature k NN is the strongest baseline and sits well above every other baseline (Table 1, clean columns). FOCUS and ZEN reach the two highest mean AUROCs, but k NN keeps the best mean rank. Our gains over the deep and TabPFN-based detectors remain large: FOCUS leads every one of them by more than four AUROC points, with statistical significance. On clean data the gain from fine-tuning is significant: FOCUS improves on ZEN on 31 of the 47 datasets ($p = 0.02$).

In practice, it is often unknown how contaminated the training set is, if at all. A practitioner who chooses FOCUS without knowing whether the clean or the contaminated regime holds therefore matches the best classical detector if the reference set turns out clean, and gains 3.6 AUROC points over the best baseline if it does not.

4 DISCUSSION

How much does each step of the feature extraction help? We measure what each part of our method contributes, in the contaminated regime. The starting point is the *plain readout*: the frozen model’s embedding of the data, read by k NN distance and nothing else (Sec. 2.2). Figure 4a ablates ZEN’s three steps, one at a time (Appendix C.4 ablates every step in both regimes, with significance tests). The plain readout reaches 74.1 AUROC, next to raw-feature k NN’ 73.9 and the best deep detector. The augmented context adds 0.9 points. The embedding with feature subsets adds 1.9 points. The soft cleaning adds 2.8 points, the largest step. Together the steps take the frozen model from 74.1 to 79.7, above every baseline.

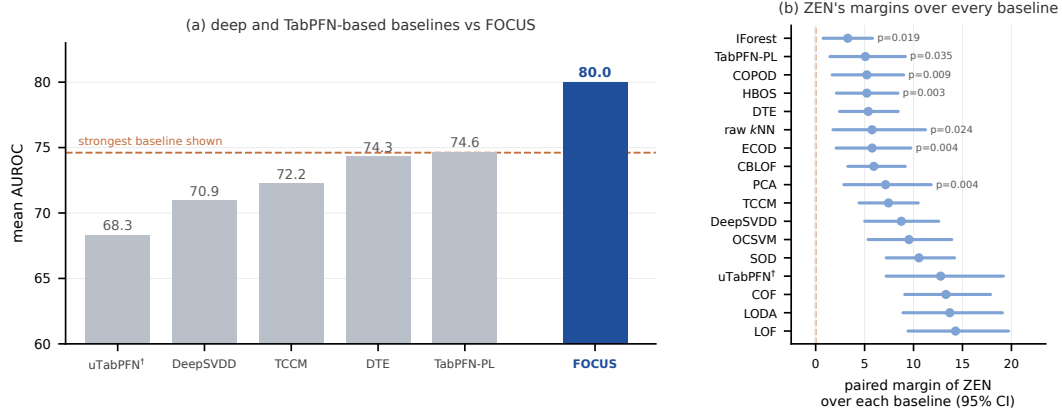


Figure 3: Additional results, contaminated regime. (a) Mean AUROC of the deep and TabPFN-based detectors and of FOCUS over the 47 datasets. Dashed line: the strongest baseline shown (TabPFN-PL). (b) Paired AUROC margin of ZEN over each baseline. Dot: mean; bar: 95% bootstrap interval over datasets. Every margin is positive and significant (Wilcoxon $p < 0.05$); the p -value is printed where it exceeds 0.001. [†]uTabPFN covers 44 of 47 datasets.

How and why is FOCUS resistant to contamination? Density-based detectors rate a sample by how far it sits from its nearest reference samples. When the reference set holds undetected anomalies, a test anomaly that is similar to them finds close neighbors and appears normal. Figure 4b shows the cost the contamination induces on each dataset: between the clean and the contaminated regime raw-feature k NN loses 11 AUROC points on average, while FOCUS loses only 5.9 points.

FOCUS limits the damage from contamination by finding the anomalies hiding in the reference set and preventing them from “hiding” the test anomalies. Its isolation estimate scores every reference sample by how far it sits from its nearest other reference samples. Used as an anomaly score for the reference samples, the estimate reaches 76.7 AUROC (Appendix C.4, Figure 9). The multi-subset averaging is essential here. It works because the runs use different features, and therefore make different mistakes during soft cleaning. The trust weights built from the pooled estimate average 0.70 on a reference anomaly and 1.16 on a normal sample, so a test anomaly next to a reference anomaly no longer looks normal. Appendix C.4 gives the full breakdown.

Does our approach transfer to other tabular foundation models? Yes. We applied ZEN, the training-free part of our approach, with the same feature subsets, blocks, constants and protocol, to four other in-context tabular models, three from other groups. TabPFN v2 (Hollmann et al., 2025), TabICL (Qu et al., 2025) and Mitra (Zhang et al., 2025) are, like TabPFN-3, prior-data fitted networks trained on synthetic priors; TabDPT (Ma et al., 2025) is pretrained on real tables instead. Figure 4c shows our improvements with these backbones in the contaminated regime. Appendix C.10 gives more details on both regimes. ZEN lifts every backbone by roughly five to eight AUROC points, and every gain is statistically significant. The FOCUS fine-tune transfers as well: on TabPFN v2 it lifts the plain readout by 3.0 AUROC points (Appendix C.10). For two of these models, TabICL and TabDPT, the results are about as strong as with TabPFN-3, our main backbone.

How sensitive are ZEN and FOCUS to their constants? Not much: no constant sits at a sharp optimum. We ablated every constant one at a time, on the same datasets, seeds and splits as the main results. Under contamination, reading the last one to four blocks of TabPFN gives a similar result, and reading more blocks, or earlier ones, is worse. $k = 50$ sits on a plateau (Table 12). Every λ between 0.25 and 2 keeps ZEN above the strongest deep baseline. Each added feature subset helps, with diminishing returns. For FOCUS, reducing the number of trainable blocks to three, for example, lowers mean AUROC by 0.5 points in both regimes. Training for one or three epochs instead of two, or keeping the last epoch instead of the average, changes it by at most 0.2 points. Finally, the gain of the soft cleaning grows with the contamination rate: when 2% of the reference set are injected anomalies the cleaning adds 0.1 AUROC points, and when 20% are, it adds 2.0. Appendix C gives the full details.

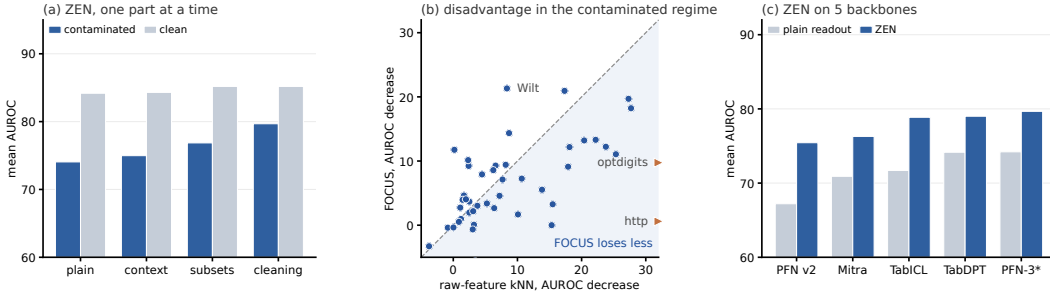


Figure 4: Ablations and analysis. (a) The ablation ladder: mean AUROC of the frozen model as ZEN’s steps are added one at a time, the plain readout, then the augmented context, the feature subsets and the soft cleaning, in both regimes. (b) The disadvantage of moving from the clean to the contaminated regime. the AUROC each method loses: FOCUS on the y -axis against raw-feature k NN on the x -axis, one point per dataset. In the shaded region FOCUS loses less due to contamination. Triangles are datasets beyond the axis. (c) ZEN’s first step, the plain readout, and full ZEN on five backbones under contamination, every constant unchanged. The asterisk marks the paper’s backbone, TabPFN-3.

5 RELATED WORK

Classical anomaly detectors. Classical tabular anomaly detection comes from several algorithm families (Zhao et al., 2019; Han et al., 2022). *Neighborhood and local-density methods* score samples relative to nearby reference points. k NN uses distance to neighboring samples, while LOF compares the local density of a sample with that of its neighbors (Breunig et al., 2000). *Partitioning, projection, and ensemble methods* avoid explicit local-density estimation. Isolation Forest isolates anomalies through recursive random partitions (Liu et al., 2008), LODA aggregates weak detectors built on random one-dimensional projections (Pevný, 2016), and feature bagging combines detectors across random feature subsets (Lazarevic & Kumar, 2005). ZEN’s feature subsets are closest to this last idea, though each subset changes the in-context representation itself. *Distributional and support-estimation methods* model the normal region more directly. HBOS estimates feature-wise densities using histograms (Goldstein & Dengel, 2012), COPOD scores samples by their extremeness in the tails of empirical distributions (Li et al., 2020), and one-class SVM estimates the support of the data in a kernel feature space (Schölkopf et al., 2001).

Deep anomaly detectors. An early self-supervised line of works brought transformation-based objectives to tabular data: GOAD classifies random affine transformations of the data (Bergman & Hoshen, 2020); NeuTraL AD learns the transformations themselves (Qiu et al., 2021); ICL trains with an internal contrastive objective (Shenkar & Wolf, 2022). DTE estimates the diffusion time of a sample and scores it accordingly (Livernoche et al., 2024). TCCM learns a flow-matching-style contraction field and scores by the deviation at one time step (Li et al., 2025). MCM scores by masked cell modeling (Yin et al., 2024). NPT-AD reconstructs masked features using a non-parametric transformer that alternates attention across features within each sample and attention across samples (Thimonier et al., 2024). Noise-evaluation methods learn to separate the data from artificially corrupted copies of it (Dai et al., 2025). DeepSVDD introduced the fixed-center compactness objective FOCUS builds on (Ruff et al., 2018). All of these methods pay a per-dataset training cost; in our pipeline the only training anywhere is FOCUS’s two-epoch fine-tune.

Pretrained anomaly detectors. A recent line of work pretrains dedicated detectors once and applies them to new datasets without further training: TACTIC (Marszałek et al., 2026) and FoMo-0D (Shen et al., 2025) are pretrained on synthetic anomaly-detection tasks for zero-shot detection. OutFormer (Ding et al., 2026) is pretrained on synthetic labeled datasets and infers test labels in context. OFA-TAD (Li et al., 2026) trains once on multiple source datasets to generalize to unseen ones. These detectors commit to their anomaly assumptions at pretraining time. Frozen TabPFN has also served as a plug-in scorer: the official unsupervised extension scores anomalies through the library’s density construction over the frozen model (Prior Labs, 2025), and OutFormer’s appendix builds a similar frozen TabPFN baseline (Ding et al., 2026). We differ from these prior attempts to use PFNs for anomaly detection: ZEN uses the embeddings of a pretrained classification model rather than its predictions, and FOCUS adapts them.

Transfer and adaptation in anomaly detection. In images, k NN on frozen pretrained features outperformed earlier self-supervised detectors (Bergman et al., 2020). Anomaly localization methods were built on the same kind of features (Cohen & Hoshen, 2020; Roth et al., 2022). PANDA, MSAD, and Transformatly adapt the pretrained features to the normal training data (Reiss et al., 2021; Reiss & Hoshen, 2023; Cohen & Avidan, 2022). Under contamination, SRR refines the training set with an ensemble of one-class classifiers as training proceeds (Yoon et al., 2022), LOE infers latent labels for the unlabeled mix during training (Qiu et al., 2022), and EPHAD adjusts the outputs of a detector trained on contaminated data with evidence gathered at test time (Patra & Ben Taieb, 2025). FOCUS applies a related idea: it soft-cleans the reference set using the given initial representation.

6 LIMITATIONS

Hardware and runtime. Scoring with ZEN or FOCUS runs TabPFN forward passes for each of the $B+1 = 6$ feature subsets. Both therefore need a GPU: on one A100, our method processes 3,500 samples per minute (Appendix D.3). Although this is slower than many classical detectors, most of that time is taken by the forward passes themselves, with ten leave-one-fold-out passes per feature subset (Ye et al., 2025). This cost can be cut in several ways. Fewer leave-one-fold-out folds cost nothing measurable and can substantially reduce computation time (Appendix C.8), and fewer feature subsets trade marginal accuracy for a significant speedup (Appendix C.6). Every speedup of the backbone’s forward pass, such as the reduced KV cache of the TabPFN-3 report (Grinsztajn et al., 2026), lowers our cost directly.

Scale. ADBench’s standard evaluation subsamples datasets above 10,000 samples (Sec. 3.1), so the largest reference set our methods have seen holds 10,000 samples. Every score is a set of TabPFN forward passes over the whole reference set, so the cost grows with its size, and larger sets are untested. Yet, TabPFN-3 handles up to one million samples (Grinsztajn et al., 2026), and future models will likely accommodate even more. Learned contexts (Feuer et al., 2024) and test-time adaptations (Ye et al., 2025), are emerging options to extend the same methods to larger tables. Three datasets in ADBench have more than 200 features. In these cases we used a fixed subset of 200 features (Appendix A.3). A more sophisticated treatment of wide tables can further improve our method.

7 CONCLUSION

In this paper we show that pretrained PFNs, once adapted to the task, make a strong anomaly detector. Our approach first embeds the data with the frozen model on random feature subsets and softly cleans the reference set. Under contamination it already reaches a higher mean AUROC than every classical and deep baseline in our comparison. FOCUS, our trust-weighted compactness fine-tune, reaches the highest mean AUROC in both clean and contaminated regimes and is significantly ahead of every baseline in the contaminated regime.

For the PFN community our results are evidence that in-context representations carry well beyond their pretraining objective. We also show that our method of extracting meaningful representations for anomaly detection, applied without any tuning or change, significantly improves four other tabular foundation models as well. None of the examined models was pretrained for anomaly detection, yet all of them exhibited strong anomaly-detection capabilities when adapted appropriately. This suggests that such capabilities are broadly accessible across in-context tabular models, even for tasks that were never used for pretraining.

AI USE STATEMENT

In this work, we used generative AI tools (LLM-based coding agents operating under continuous author direction) for tasks with required disclosure: implementing the methods and experiment infrastructure specified by the authors, assisting in the interpretation of results, and assisting with translation. We have not used generative AI tools to propose or refine hypotheses, to design or provide feedback on the research methodology or experiments, to generate synthetic datasets, to clean and reformat datasets, or to support qualitative or thematic data analysis; the methods and the experimental design are due to the authors. The remaining required-disclosure tasks, developing theoretical models or conceptual frameworks, formulating mathematical claims, providing ingredients for proving them, and writing proofs, are not applicable to this work. Additionally, we used generative AI tools for tasks with recommended disclosure: creating and editing software code, creating scientific figures, editing the text of this paper following the authors’ outlines and directions, searching and summarizing related literature, and formatting references. We have reviewed all AI-assisted work: experiment code was validated against logged design decisions before any of its results entered the paper, every reported number was recomputed from stored per-run artifacts, all text was reviewed and edited by the authors, and every citation was checked against its source. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

Every experiment in this paper runs on five fixed seeds, and every (method, dataset, seed) result is stored as an individual file; each reported number is an aggregate of those files. Section 3.1 summarizes and Appendix A specifies the datasets, the two regimes, the preprocessing, and the evaluation protocol, including the data-leak correction to the original generator. Appendix D.1 lists the complete configuration: the backbone version, the readout layers, the fine-tuning schedule, and every constant, and Appendix B gives per-dataset tables for all methods. We release the code needed to reproduce every result in this paper in a public repository (<https://github.com/Maximilianbl/zen-focus-tabular-anomaly-detection>).

ETHICS STATEMENT

This work uses only publicly available, anonymized benchmark datasets and involves no human subjects or personally identifiable information. The contribution is methodological. Anomaly detectors do affect people once deployed, in fraud screening or medical triage among others, and any such deployment should audit error rates across the populations it touches; nothing in our method removes that obligation. We are aware of no other ethical concerns.

REFERENCES

- Alessio Benavoli, Giorgio Corani, and Francesca Mangili. Should we really use post-hoc tests based on mean-ranks? *Journal of Machine Learning Research*, 17(5):1–10, 2016.
- Liron Bergman and Yedid Hoshen. Classification-based anomaly detection for general data. In *International Conference on Learning Representations (ICLR)*, 2020.
- Liron Bergman, Niv Cohen, and Yedid Hoshen. Deep nearest neighbor anomaly detection. *arXiv preprint arXiv:2002.10445*, 2020.
- Rishi Bommasani, Drew A. Hudson, Ehsan Adeli, et al. On the opportunities and risks of foundation models. *arXiv preprint arXiv:2108.07258*, 2021.
- Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.
- Markus M. Breunig, Hans-Peter Kriegel, Raymond T. Ng, and Jörg Sander. Lof: Identifying density-based local outliers. In Weidong Chen, Jeffrey F. Naughton, and Philip A. Bernstein (eds.), *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data, May 16-18, 2000, Dallas, Texas, USA*, pp. 93–104. ACM, 2000. ISBN 1-58113-218-2. doi: 10.1145/342009.335388. URL <http://doi.acm.org/10.1145/342009.335388>.

- Matan Jacob Cohen and Shai Avidan. Transformaly – two (feature spaces) are better than one. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 4059–4068, 2022.
- Niv Cohen and Yedid Hoshen. Sub-image anomaly detection with deep pyramid correspondences. *arXiv preprint arXiv:2005.02357*, 2020.
- Wei Dai, Kai Hwang, and Jicong Fan. Unsupervised anomaly detection for tabular data using deep noise evaluation. In *AAAI Conference on Artificial Intelligence (AAAI)*, pp. 11553–11562, 2025.
- Andrea Dal Pozzolo, Giacomo Boracchi, Olivier Caelen, Cesare Alippi, and Gianluca Bontempi. Credit card fraud detection: A realistic modeling and a novel learning strategy. *IEEE Transactions on Neural Networks and Learning Systems*, 29(8):3784–3797, 2018. doi: 10.1109/TNNLS.2017.2736643.
- Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7:1–30, 2006.
- Xueying Ding, Haomin Wen, Simon Klüttermann, and Leman Akoglu. From zero to hero: Advancing zero-shot foundation models for tabular outlier detection. In *International Conference on Machine Learning (ICML)*, 2026.
- Elizabeth D. Dolan and Jorge J. Moré. Benchmarking optimization software with performance profiles. *Mathematical Programming*, 91(2):201–213, 2002.
- Benjamin Feuer, Robin Tibor Schirrmeister, Valeriia Cherepanova, Chinmay Hegde, Frank Hutter, Micah Goldblum, Niv Cohen, and Colin White. TuneTables: Context optimization for scalable prior-data fitted networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Markus Goldstein and Andreas Dengel. Histogram-based outlier score (hbos): A fast unsupervised anomaly detection algorithm. *KI-2012: poster and demo track*, 1:59–63, 2012.
- Markus Goldstein and Seiichi Uchida. A comparative evaluation of unsupervised anomaly detection algorithms for multivariate data. *PLoS ONE*, 11(4):e0152173, 2016. doi: 10.1371/journal.pone.0152173.
- Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.
- Léo Grinsztajn, Klemens Flöge, Oscar Key, Felix Birkel, Philipp Jund, Brendan Roof, Mihir Manium, Shi Bin Hoo, Magnus Bühler, Anurag Garg, Dominik Safaric, Jake Robertson, Benjamin Jäger, Simone Alessi, Adrian Hayler, Vladyslav Moroshan, Lennart Purucker, Philipp Singer, Alan Arazi, Julien Siems, Jan Hendrik Metzen, Georg Grab, Nick Erickson, Siyuan Guo, Elliott Kalfon, Simon Bing, David Salinas, Clara Cornu, Lilly Charlotte Wehrhahn, Diana Kriuchkova, Kursat Kaya, Lydia Sidhoum, Marie Salmon, Jerry Chen, Madelon Hulsebos, Yann LeCun, Samuel Müller, Bernhard Schölkopf, Sauraj Gambhir, Noah Hollmann, and Frank Hutter. TabPFN-3: Technical report. *arXiv preprint arXiv:2605.13986*, 2026.
- Songqiao Han, Xiyang Hu, Hailiang Huang, Minqi Jiang, and Yue Zhao. ADBench: Anomaly detection benchmark. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2022.
- Noah Hollmann, Samuel Müller, Katharina Eggensperger, and Frank Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations (ICLR)*, 2023.
- Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. doi: 10.1038/s41586-024-08328-6.

- Ali Ismail-Fawaz, Angus Dempster, Chang Wei Tan, Matthieu Herrmann, Lynn Miller, Daniel F. Schmidt, Stefano Berretti, Jonathan Weber, Maxime Devanne, Germain Forestier, and Geoffrey I. Webb. An approach to multiple comparison benchmark evaluations that is stable under manipulation of the compare set. *arXiv preprint arXiv:2305.11921*, 2023.
- Pavel Izmailov, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, pp. 876–885, 2018.
- Aleksandar Lazarevic and Vipin Kumar. Feature bagging for outlier detection. In *Proceedings of the Eleventh ACM SIGKDD International Conference on Knowledge Discovery in Data Mining (KDD)*, pp. 157–166. ACM, 2005. doi: 10.1145/1081870.1081891.
- Shiyuan Li, Yixin Liu, Yu Zheng, Xiaofeng Cao, Shirui Pan, and Heng Tao Shen. Towards one-for-all anomaly detection for tabular data. In *International Conference on Machine Learning (ICML)*, 2026.
- Zheng Li, Yue Zhao, Nicola Botta, Cezar Ionescu, and Xiyang Hu. COPOD: copula-based outlier detection. In *IEEE International Conference on Data Mining (ICDM)*. IEEE, 2020.
- Zhong Li, Qi Huang, Yuxuan Zhu, Lincen Yang, Mohammad Mohammadi Amiri, Niki van Stein, and Matthijs van Leeuwen. Scalable, explainable and provably robust anomaly detection with one-step flow matching. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou. Isolation forest. In *IEEE International Conference on Data Mining (ICDM)*, pp. 413–422, 2008.
- Victor Livernoche, Vineet Jain, Yashar Hezaveh, and Siamak Ravanbakhsh. On diffusion modeling for anomaly detection. In *International Conference on Learning Representations (ICLR)*, 2024.
- Junwei Ma, Valentin Thomas, Rasa Hosseinzadeh, Alex Labach, Hamidreza Kamkari, Jesse C. Cresswell, Keyvan Golestan, Guangwei Yu, Anthony L. Caterini, and Maksims Volkovs. TabDPT: Scaling tabular foundation models on real data. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Patryk Marszałek, Tomasz Kuśmierczyk, and Marek Śmieja. TACTIC for navigating the unknown: Tabular anomaly detection via in-context inference. *arXiv preprint arXiv:2603.14171*, 2026.
- Matthew Middlehurst, Patrick Schäfer, and Anthony Bagnall. Bake off redux: a review and experimental evaluation of recent time series classification algorithms. *Data Mining and Knowledge Discovery*, 38(4):1958–2031, 2024. doi: 10.1007/s10618-024-01022-1.
- Sukanya Patra and Souhaib Ben Taieb. An evidence-based post-hoc adjustment framework for anomaly detection under data contamination. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Tomáš Pevný. Loda: Lightweight on-line detector of anomalies. *Machine Learning*, 102(2):275–304, 2016.
- Prior Labs. TabPFN extensions. <https://github.com/PriorLabs/tabpfn-extensions>, 2025.
- Chen Qiu, Timo Pfroemer, Marius Kloft, Stephan Mandt, and Maja Rudolph. Neural transformation learning for deep anomaly detection beyond images. In *International Conference on Machine Learning (ICML)*, pp. 8703–8714, 2021.
- Chen Qiu, Aodong Li, Marius Kloft, Maja Rudolph, and Stephan Mandt. Latent outlier exposure for anomaly detection with contaminated data. In *International Conference on Machine Learning (ICML)*, pp. 18153–18167, 2022.
- Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. TabICL: A tabular foundation model for in-context learning on large data. In *International Conference on Machine Learning (ICML)*, 2025.

- Jingang Qu, David Holzmüller, Gaël Varoquaux, and Marine Le Morvan. TabICLv2: A better, faster, scalable, and open tabular foundation model. *arXiv preprint arXiv:2602.11139*, 2026.
- Tal Reiss and Yedid Hoshen. Mean-shifted contrastive loss for anomaly detection. In *AAAI Conference on Artificial Intelligence (AAAI)*, pp. 2155–2162, 2023.
- Tal Reiss, Niv Cohen, Liron Bergman, and Yedid Hoshen. PANDA: Adapting pretrained features for anomaly detection and segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2805–2813, 2021.
- Karsten Roth, Latha Pemula, Joaquin Zepeda, Bernhard Schölkopf, Thomas Brox, and Peter Gehler. Towards total recall in industrial anomaly detection. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 14298–14308, 2022.
- Lukas Ruff, Robert Vandermeulen, Nico Gönitz, Lucas Deecke, Shoaib Ahmed Siddiqui, Alexander Binder, Emmanuel Müller, and Marius Kloft. Deep one-class classification. In *International Conference on Machine Learning (ICML)*, pp. 4393–4402, 2018.
- Bernhard Schölkopf, John C Platt, John Shawe-Taylor, Alex J Smola, and Robert C Williamson. Estimating the support of a high-dimensional distribution. *Neural computation*, 13(7):1443–1471, 2001.
- Yuchen Shen, Haomin Wen, and Leman Akoglu. FoMo-0D: A foundation model for zero-shot tabular outlier detection. *Transactions on Machine Learning Research (TMLR)*, 2025.
- Tom Shenkar and Lior Wolf. Anomaly detection for tabular data with internal contrastive learning. In *International Conference on Learning Representations (ICLR)*, 2022.
- Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90, 2022.
- Hugo Thimonier, Fabrice Popineau, Arpad Rimmel, and Bich-Liên Doan. Beyond individual input for deep anomaly detection on tabular data. In *International Conference on Machine Learning (ICML)*, pp. 48097–48123, 2024.
- Han-Jia Ye, Si-Yang Liu, and Wei-Lun Chao. A closer look at TabPFN v2: Understanding its strengths and extending its capabilities. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Jiaxin Yin, Yuanyuan Qiao, Zitang Zhou, Xiangchao Wang, and Jie Yang. MCM: Masked cell modeling for anomaly detection in tabular data. In *International Conference on Learning Representations (ICLR)*, 2024.
- Jinsung Yoon, Kihyuk Sohn, Chun-Liang Li, Serkan O. Arik, Chen-Yu Lee, and Tomas Pfister. Self-supervise, refine, repeat: Improving unsupervised anomaly detection. *Transactions on Machine Learning Research (TMLR)*, 2022.
- Sanghyu Yoon, Dongmin Kim, Suhee Yoon, Ye Seul Sim, Seungdong Yoa, Hye-Seung Cho, Soonyoung Lee, Hankook Lee, and Woohyung Lim. ReTabAD: A benchmark for restoring semantic context in tabular anomaly detection. In *International Conference on Learning Representations (ICLR)*, 2026.
- Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. How transferable are features in deep neural networks? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
- Xiyuan Zhang, Danielle C. Maddix, Junming Yin, Nick Erickson, Abdul Fatir Ansari, Boran Han, Shuai Zhang, Leman Akoglu, Christos Faloutsos, Michael W. Mahoney, Cuixiong Hu, Huzefa Rangwala, George Karypis, and Bernie Wang. Mitra: Mixed synthetic priors for enhancing tabular foundation models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
- Yue Zhao, Zain Nasrullah, and Zheng Li. PyOD: A Python toolbox for scalable outlier detection. *Journal of Machine Learning Research*, 20(96):1–7, 2019.

Table 2: The twelve datasets the official resampling affects, with the fraction of test samples whose byte-identical twin lands in the training set under that protocol (mean over seeds 0–4). Our protocol runs these datasets at their natural size, where the fraction is zero.

Dataset	n	clean: % test normals	contam.: % test samples
breastw	683	68.2	59.9
glass	214	96.0	95.1
Hepatitis	80	100.0	100.0
Ionosphere	351	93.2	87.2
Lymphography	148	99.8	99.5
Pima	768	72.0	58.1
Stamps	340	91.4	88.7
vertebral	240	96.8	94.6
WBC	223	95.8	96.4
WDBC	367	86.4	85.7
wine	129	99.4	99.4
WPBC	198	99.4	97.1
<i>Mean</i>		91.5	88.5

A EXPERIMENTAL PROTOCOL IN FULL

A.1 SPLITS AND SUBSAMPLING

This appendix completes the protocol of Sec. 3.1. In the clean regime the normal-class indices are shuffled, and the first $\lfloor 0.7n \rfloor$ of them form the training set, where n is the dataset’s size after subsampling. The test set holds every remaining normal sample and every anomaly. When a dataset has too few normal samples to fill that count, the fraction steps down by 0.1 until it fits; this keeps the split arithmetic of the standard pipeline. Eight datasets with anomaly rates above 30% step down to 0.6. On the most contaminated of them the clean test set is left with few normal samples; SpamBase (39.9% anomalies) keeps four. Every method uses the same split, so no comparison is affected. In the contaminated regime the split is stratified 70/30 and preserves the anomaly rate on both sides.

Datasets above 10,000 samples are subsampled to exactly 10,000 before splitting. The draw is without replacement, separately from the normal and the anomaly pool: the anomaly count is $\text{round}(\text{natural rate} \times 10,000)$, floored at two and capped at the true anomaly count, and normal samples fill the rest. This preserves the natural anomaly rate exactly rather than in expectation, and it can never duplicate a sample.

A.2 THE RESAMPLING LEAK IN THE OFFICIAL ADBENCH PIPELINE

ADBench’s official generator resamples every dataset below 1,000 samples to exactly 1,000 by drawing samples with replacement, before the train/test split (Han et al., 2022). Copies of one physical sample can therefore land on both sides of the split. Twelve of the 47 datasets are affected. Table 2 reports, for each of them, the fraction of test samples whose byte-identical twin sits in the training set under that protocol, measured by replicating the official code path on the real data over seeds 0–4. Natural duplicate samples in the raw data are not counted; without the resampling step the fraction is structurally zero. In the contaminated regime, on average 88.5% of all test samples have a training twin. In the clean regime the training set is normal-only, so anomalies cannot leak; instead 91.5% of the test *normal* samples have one. For a distance- or density-based scorer a twin sits at distance zero, which drives the affected normal test scores toward their minimum and widens the apparent normal/anomaly separation. The inflation therefore acts upward on exactly the detector families these benchmarks rank.

We evaluate the affected datasets at their natural size instead: no resampling, no duplication. The 10,000-sample subsample of Sec. 3.1 is unrelated to this leak; it draws without replacement and duplicates nothing. One consequence is deliberate: our numbers are not comparable cell-for-cell with numbers produced under the official pipeline, on these twelve datasets by construction and in any aggregate through them.

Table 3: The 47 ADBench classical datasets: natural size n , dimensionality d , and anomaly rate. [†] below 1,000 samples (affected by the official pipeline’s resampling; run at natural size here). [‡] subsampled to 10,000 samples. [§] ZEN and FOCUS use a fixed 200-feature subset.

Dataset	n	d	anom. %	Dataset	n	d	anom. %
ALOI [†]	49,534	27	3.0	musk	3,062	166	3.2
annthyroid	7,200	6	7.4	optdigits	5,216	64	2.9
backdoor [‡]	95,329	196	2.4	PageBlocks	5,393	10	9.5
breastw [†]	683	9	35.0	pendigits	6,870	16	2.3
campaign [‡]	41,188	62	11.3	Pima [†]	768	8	34.9
cardio	1,831	21	9.6	satellite	6,435	36	31.6
Cardiotocography	2,114	21	22.0	satimage-2	5,803	36	1.2
celeba [‡]	202,599	39	2.2	shuttle [‡]	49,097	9	7.2
census ^{‡§}	299,285	500	6.2	skin [‡]	245,057	3	20.8
cover [‡]	286,048	10	0.96	smtp [‡]	95,156	3	0.03
donors [‡]	619,326	10	5.9	SpamBase	4,207	57	39.9
fault	1,941	27	34.7	speech [§]	3,686	400	1.6
fraud [‡]	284,807	29	0.17	Stamps [†]	340	9	9.1
glass [†]	214	7	4.2	thyroid	3,772	6	2.5
Hepatitis [†]	80	19	16.2	vertebral [†]	240	6	12.5
http [‡]	567,498	3	0.39	vowels	1,456	12	3.4
InternetAds [§]	1,966	1555	18.7	Waveform	3,443	21	2.9
Ionosphere [†]	351	32	35.9	WBC [†]	223	9	4.5
landsat	6,435	36	20.7	WDBC [†]	367	30	2.7
letter	1,600	32	6.2	Wilt	4,819	5	5.3
Lymphography [†]	148	18	4.1	wine [†]	129	13	7.8
magic.gamma [‡]	19,020	10	35.2	WPBC [†]	198	33	23.7
mammography [‡]	11,183	6	2.3	yeast	1,484	8	34.2
mnist	7,603	100	9.2				

A.3 NOTES ON SPECIFIC DATASETS AND DETECTORS

TabPFN-3 is characterized up to 200 input features (Grinsztajn et al., 2026), and past that the library switches to an internal ensemble of estimators. Our fine-tuning implementation trains a single shared module, so that ensemble would multiply the cost and break the design. On the three datasets with more than 200 features (census, InternetAds, speech), ZEN and FOCUS therefore use a fixed subset of 200 features, drawn once with constant features excluded; every other method uses every feature. uTabPFN builds its density one feature at a time, so its cost grows with the feature count. On the same three datasets a single seed takes about five to seven GPU-hours, and we did not run all five. Its aggregates therefore cover 44 of 47 datasets, marked wherever they appear. Four PyOD detectors (COPOD, ECOD, SOD, and COF) compute part of their statistics over the batch being scored, which is how the library and ADBench run them; every other method, ours included, scores each test sample from training-set state alone. CBLOF fails on a small number of seeds when its clustering cannot separate the data; its dataset means average the completed seeds.

B DATASETS AND PER-DATASET RESULTS

B.1 DATASET STATISTICS

Table 3 lists the 47 datasets with their natural size n , dimensionality d , and anomaly rate, computed from the benchmark files our loader reads. The markers flag the three protocol facts that touch individual datasets: the sub-1,000 sets the official resampling would affect (Appendix A.2), the sets subsampled to 10,000 samples, and the three sets where ZEN and FOCUS use a fixed 200-feature subset (Appendix A.3).

B.2 PER-DATASET AUROC

Tables 4 through 7 give every compared method’s per-dataset AUROC, averaged over the five seeds. The bottom rows reproduce the means of Table 1.

B.3 AUPRC

Table 8 reports each method’s mean AUPRC (average precision, $\times 100$) and its mean per-dataset AUPRC rank. AUPRC depends strongly on the test anomaly rate, which spans 0.03% to 39.9% across the contaminated regime’s datasets (Table 3) and runs higher still in the clean regime, whose test sets hold every anomaly. A cross-dataset mean therefore weights datasets very unevenly, and the rank column is the more comparable summary. Every value is computed from stored score vectors, and a method’s AUPRC is reported only where those vectors reproduce the method’s official AUROC. FOCUS has the best mean AUPRC in both regimes and the best mean rank under contamination. Its contaminated lead over Isolation Forest is not significant ($p = 0.08$), while its lead over raw-feature k NN is ($p = 0.002$). In the clean regime raw-feature k NN keeps the best AUPRC rank, and FOCUS’s lead over it is not significant ($p = 0.52$). FOCUS improves on ZEN’s AUPRC in both regimes, significantly on clean data ($p = 0.03$) and not under contamination ($p = 0.15$).

B.4 SEED VARIABILITY

Every number in Tables 1 and 4–7 is a mean over five seeds, each seed drawing a fresh split. Table 9 reports the spread behind those means: the standard deviation of each method’s 47-dataset mean across the five seeds, and the median over datasets of the per-dataset standard deviation across seeds. The seed-to-seed spread of a 47-dataset mean is below one AUROC point for 14 of the 18 methods with a reported spread under contamination and for 17 of the 19 on clean data, and below 1.3 for every method but DeepSVDD; the per-dataset spread does not track dataset size.

C STATISTICAL COMPARISONS

C.1 CONSTRUCTION OF THE MAIN COMPARISON (FIGURE 2)

Figure 2 compares FOCUS with the baselines in the contaminated regime, following the procedure of Demšar (2006) for comparing one proposed method with several others. Methods are ranked within each dataset on their five-seed mean AUROC, rank 1 the best, and the axis of the figure is the mean rank over the 47 datasets. The ranked pool holds FOCUS and the 16 baselines that cover all 47 datasets; uTabPFN covers 44. ZEN, our own frozen detector, is not a baseline; its comparison with FOCUS is in Sec. 3. A Friedman test over the 17 methods rejects the hypothesis that all perform alike ($p < 10^{-13}$). FOCUS is then tested against each of the 16 baselines with a two-sided Wilcoxon signed-rank test on the 47 paired values (Benavoli et al., 2016), and the 16 p -values are Holm-corrected. All 16 are significant; the largest Holm-adjusted p -value is 0.034, against raw-feature k NN and TabPFN-PL. The bars among the baselines are critical-difference cliques: Wilcoxon tests over all 120 of their pairs, Holm-corrected, with a bar joining each maximal run of adjacent methods among which no pair differs significantly. Against uTabPFN, on its 44 datasets, FOCUS is ahead by 13.1 AUROC points ($p < 0.001$). FOCUS beats Isolation Forest on 28 of the 47 datasets; the signed-rank test weighs the size of each difference, and FOCUS’s wins are the larger ones.

C.2 PER-REGIME DIAGRAMS AND THE PAIRWISE MATRIX

Figure 2 tests FOCUS against the baselines in the contaminated regime; this subsection ranks every method in each regime. Because rank-based summaries depend on which methods are in the ranked pool (Benavoli et al., 2016), Figure 2 draws its significance from pairwise per-dataset tests; this subsection reports the all-pairs rank-based comparisons. Figure 5 gives the Holm-corrected critical-difference diagram of each regime separately, over every ranked method including ZEN and FOCUS. Figure 6 reports the complete pairwise picture for the contaminated regime: the mean AUROC difference, the win/tie/loss count, and the Wilcoxon p -value of our methods against every baseline, with each p -value reported as computed, before any multiple-comparison correction.

Table 4: Per-dataset AUROC ($\times 100$, five-seed means), contaminated regime. Methods ordered by mean AUROC; ours marked, markers as in Table 1. The 19 methods are split over two tables for width: this table holds the stronger half, Table 5 the rest, same 47 datasets.

Dataset	FOCUS (ours)	ZEN (ours)	IForest	TabPFN-PL [†]	COPOD	HBOS	DTE ^d	raw k-NN	ECOD	CBLOF
ALOI	56.0	55.5	53.9	61.5	52.3	54.6	55.8	59.0	54.7	56.1
annthyroid	85.9	84.3	82.9	68.7	78.0	62.9	65.1	72.3	79.1	65.6
backdoor	89.1	87.6	72.9	91.7	78.5	70.3	90.2	76.6	84.2	77.4
breastw	98.1	97.7	98.9	97.7	99.4	98.6	89.9	98.4	99.1	97.1
campaign	77.4	78.6	71.2	80.6	78.7	79.3	72.2	69.1	77.5	65.9
cardio	90.0	86.9	93.1	95.0	91.9	84.6	80.3	80.8	94.2	85.4
Cardiotocography	62.6	62.8	68.8	65.6	65.7	59.8	60.1	55.0	78.5	62.9
celeba	76.2	71.7	69.6	73.9	74.7	75.1	68.2	62.4	75.2	61.1
census	63.2	62.4	59.9	60.9	66.6	62.8	65.5	64.1	65.7	58.5
cover	89.4	91.8	89.6	93.3	90.2	78.5	93.6	90.2	93.4	90.0
donors	87.5	86.8	77.6	68.6	82.5	78.7	74.3	81.8	89.4	74.0
fault	71.0	70.1	56.7	54.4	46.6	56.6	60.9	73.8	47.8	65.2
fraud	95.6	95.9	94.4	93.4	93.7	95.0	94.0	95.8	94.0	93.0
glass	84.3	85.4	80.2	75.3	72.9	81.4	70.5	87.0	68.8	74.5
Hepatitis	75.0	76.2	76.5	84.2	82.2	77.5	75.2	74.8	77.2	65.8
http	99.3	99.4	100.0	91.5	99.0	99.5	99.5	1.6	98.0	99.5
InternetAds	67.6	67.7	68.0	70.3	67.3	68.7	61.8	71.9	67.4	69.7
Ionosphere	89.5	89.6	81.9	88.4	77.2	61.5	95.3	90.0	70.8	86.4
landsat	55.5	58.0	49.3	47.3	41.2	56.5	47.8	59.3	35.7	55.8
letter	87.4	87.6	63.4	64.1	55.5	60.2	74.8	86.8	56.3	73.8
Lymphography	99.5	99.5	100.0	96.5	100.0	99.8	99.5	100.0	99.5	100.0
magic.gamma	75.6	77.9	72.8	65.5	68.0	70.4	76.0	83.0	63.7	76.0
mammography	84.8	84.5	85.8	82.2	90.3	84.8	82.2	84.6	90.4	81.8
mnist	86.3	84.6	78.8	87.5	77.7	61.8	77.2	82.8	75.1	79.7
musk	100.0	100.0	99.8	99.4	94.1	100.0	84.0	84.7	95.0	100.0
optdigits	89.5	92.7	75.4	36.5	67.6	82.1	54.4	39.9	59.5	83.4
PageBlocks	96.5	94.9	90.1	89.1	87.8	80.5	90.4	85.2	91.8	87.1
pendigits	88.4	88.4	93.2	92.5	90.7	92.8	79.9	74.6	92.7	96.5
Pima	68.8	69.7	70.0	64.6	64.8	68.3	63.6	73.6	60.3	67.6
satellite	71.4	67.9	70.0	68.8	62.5	75.4	68.3	67.3	57.4	71.0
satimage-2	99.8	99.7	99.4	95.5	97.3	97.8	99.2	96.7	96.1	99.9
shuttle	80.3	72.5	99.7	99.4	99.6	99.0	98.0	72.4	99.3	74.0
skin	79.5	74.2	67.2	40.7	46.6	58.8	80.5	71.0	48.4	63.6
smtp	84.8	83.7	87.3	72.0	87.7	82.5	83.1	89.9	82.5	82.4
SpamBase	69.7	76.6	62.6	40.8	69.2	65.0	54.6	54.8	65.8	56.5
speech	48.4	47.6	49.5	48.3	49.3	46.8	51.9	48.9	46.8	46.6
Stamps	88.2	88.5	90.0	87.9	93.2	90.6	78.0	89.3	86.7	72.6
thyroid	98.3	98.0	98.0	95.7	94.1	95.5	95.0	94.9	97.9	92.8
vertebral	32.7	32.0	37.8	41.9	36.4	30.9	42.7	35.0	45.3	39.3
vowels	96.0	97.9	70.0	74.1	42.6	64.6	83.3	97.0	51.0	89.9
Waveform	79.2	79.0	65.9	73.2	70.8	65.4	61.7	74.3	56.5	69.7
WBC	99.6	99.6	99.5	99.3	99.4	99.2	97.2	99.4	99.4	98.6
WDBC	96.0	95.9	98.1	97.4	99.1	99.2	93.6	97.0	96.5	97.7
Wilt	69.1	64.7	46.4	44.1	34.2	38.3	31.7	45.7	39.9	33.0
wine	85.0	85.2	86.5	94.4	92.4	90.7	78.1	88.5	81.5	30.7
WPBC	49.3	48.4	49.9	51.6	53.4	56.4	51.8	55.3	44.3	53.6
yeast	43.5	45.1	38.2	41.3	36.2	40.1	40.9	37.6	43.4	43.8
<i>Mean</i>	80.0	79.7	76.4	74.6	74.4	74.4	74.3	73.9	73.9	73.7

Table 5: Per-dataset AUROC ($\times 100$, five-seed means), contaminated regime: the remaining methods, continuing Table 4 over the same 47 datasets. [†]uTabPFN covers 44 of 47 datasets.

Dataset	PCA	TCCM ^d	DeepSVDD ^d	OCSVM	SOD	uTabPFN ^{††}	COF	LODA	LOF
ALOI	56.5	57.3	56.2	55.1	61.0	61.0	66.1	50.4	66.6
annthyroid	66.1	74.3	69.3	59.1	79.3	87.2	70.7	44.4	71.0
backdoor	81.0	90.7	78.6	83.3	69.8	41.1	72.1	76.3	76.9
breastw	94.6	91.2	91.5	87.2	94.0	76.8	44.5	99.1	46.8
campaign	74.6	73.0	70.4	67.8	70.3	50.3	56.3	57.2	57.3
cardio	95.0	69.5	89.1	93.9	67.0	56.6	60.8	88.0	67.4
Cardiotocography	74.8	78.9	73.1	78.1	51.8	52.0	55.3	75.0	57.4
celeba	77.6	67.6	71.7	68.8	50.1	43.8	43.1	59.2	42.2
census	65.7	59.0	65.3	53.4	60.1	–	42.1	44.3	47.4
cover	94.1	93.2	85.6	92.9	74.9	66.5	80.6	89.1	86.6
donors	82.7	80.0	79.1	72.3	62.2	63.2	70.8	47.3	59.0
fault	49.3	61.2	51.9	53.0	66.7	56.3	60.7	48.9	58.7
fraud	94.8	94.2	93.1	94.1	95.2	95.9	93.5	93.9	95.7
glass	69.8	76.9	60.2	36.9	84.4	78.1	85.3	71.8	80.9
Hepatitis	78.0	82.5	77.8	71.5	62.7	74.5	70.0	61.3	74.5
http	99.7	84.8	99.8	99.5	39.0	97.0	92.7	6.6	26.8
InternetAds	60.4	68.7	59.2	68.9	61.8	–	64.6	58.6	64.0
Ionosphere	77.0	86.3	72.8	73.7	85.1	74.2	85.2	77.3	87.6
landsat	35.1	52.8	38.4	34.9	60.9	56.4	53.5	37.0	55.0
letter	51.2	82.1	50.3	48.1	86.3	30.9	80.9	52.0	85.4
Lymphography	99.5	99.8	99.1	99.8	95.6	5.8	97.7	68.8	99.5
magic.gamma	66.9	70.8	63.5	60.1	76.7	84.1	66.5	68.2	68.4
mammography	89.9	82.3	81.6	86.6	79.6	85.0	76.7	83.7	76.3
mnist	85.2	74.0	70.8	83.0	58.3	61.0	70.6	74.2	68.8
musk	100.0	70.8	96.0	92.4	76.5	96.2	44.5	95.0	35.8
optdigits	51.3	60.6	51.8	55.0	57.1	84.0	51.3	55.7	56.3
PageBlocks	91.3	79.5	90.2	90.1	78.2	77.5	72.5	83.4	72.9
pendigits	93.6	75.5	85.5	94.9	65.1	79.1	44.7	91.4	42.6
Pima	65.1	68.7	64.0	61.8	57.4	57.2	65.0	63.7	63.6
satellite	58.9	63.5	60.8	58.9	64.7	69.7	54.9	60.5	54.5
satimage-2	97.5	87.9	97.2	98.1	86.7	95.8	57.7	98.2	54.9
shuttle	99.0	58.2	98.7	98.2	69.2	97.5	52.4	81.6	52.8
skin	12.2	65.2	47.8	48.5	59.0	69.8	41.7	42.9	44.9
smtp	72.6	69.3	72.7	69.3	58.5	82.3	74.2	68.1	93.7
SpamBase	54.9	47.7	54.0	52.3	52.3	61.5	41.4	43.8	44.5
speech	46.7	50.0	54.0	46.0	50.2	–	45.8	47.8	49.0
Stamps	90.6	72.1	90.8	82.9	80.9	74.0	81.0	88.4	69.6
thyroid	95.8	94.0	96.4	86.8	91.6	96.7	87.3	78.0	85.7
vertebral	44.7	44.4	37.8	42.9	40.1	35.8	41.0	30.7	37.0
vowels	51.2	91.0	53.2	48.1	90.8	90.6	95.9	64.2	93.4
Waveform	60.1	51.6	54.4	51.5	63.4	64.8	69.6	56.4	71.8
WBC	98.8	98.6	98.1	99.5	97.0	44.6	99.5	99.0	92.1
WDBC	98.0	97.7	96.2	97.5	94.6	98.1	96.3	97.4	97.2
Wilt	37.7	31.0	34.6	36.1	60.6	69.7	55.0	34.8	52.4
wine	85.2	73.3	65.7	78.1	65.9	66.7	90.0	88.9	91.5
WPBC	46.5	45.6	45.7	45.4	51.5	58.1	53.4	51.6	52.9
yeast	38.1	47.7	39.7	40.7	45.2	37.5	43.8	46.3	44.3
<i>Mean</i>	72.5	72.2	70.9	70.1	69.1	68.3	66.4	66.0	65.4

Table 6: Per-dataset AUROC ($\times 100$, five-seed means), clean regime. Methods ordered by mean AUROC; ours marked, markers as in Table 1. The 19 methods are split over two tables for width: this table holds the stronger half, Table 7 the rest, same 47 datasets.

Dataset	FOCUS (ours)	ZEN (ours)	raw kNN	LOF	CBLOF	DTE ^d	TCCM ^d	IForest	TabPFN-PL ^t	HBOS
ALOI	67.7	60.0	59.2	66.8	55.2	55.1	56.4	54.5	61.4	53.6
annthyroid	91.5	90.2	86.2	91.5	73.8	73.4	83.6	91.2	82.6	70.8
backdoor	98.2	97.4	94.5	93.4	83.1	92.4	92.9	76.7	86.7	65.7
breastw	99.1	99.6	99.6	79.0	99.5	98.5	98.3	99.8	99.3	99.6
campaign	77.9	79.1	70.0	64.8	68.9	74.9	75.1	73.5	82.5	80.0
cardio	93.2	91.3	96.3	94.9	95.9	96.4	95.4	95.1	94.0	87.3
Cardiotocography	74.8	71.4	78.8	79.9	79.0	78.4	82.6	79.9	80.4	70.7
celeba	67.1	67.6	66.7	45.6	69.2	73.5	70.5	71.4	77.5	77.7
census	65.2	64.6	66.6	49.4	62.2	68.2	60.4	59.1	65.7	65.0
cover	98.7	98.1	96.8	98.8	94.0	97.9	92.8	85.1	90.8	77.4
donors	99.7	99.2	100.0	95.6	93.1	96.4	91.9	90.2	90.4	82.8
fault	79.6	75.5	80.0	64.3	71.2	70.6	70.4	65.0	67.2	64.0
fraud	92.3	92.7	92.1	93.9	89.7	89.9	90.2	89.3	88.8	89.5
glass	88.9	85.3	88.7	73.0	89.0	74.3	69.9	83.1	85.7	81.9
Hepatitis	79.6	81.8	82.0	80.4	71.6	77.2	86.9	79.6	85.5	82.5
http	99.9	99.8	99.9	99.9	99.5	99.5	100.0	99.2	91.6	99.3
InternetAds	70.3	66.7	78.3	86.0	72.6	80.0	70.7	39.2	83.2	54.5
Ionosphere	96.6	96.2	97.7	94.7	98.6	97.6	97.6	92.0	82.0	77.8
landsat	76.5	76.2	76.7	76.2	69.2	51.1	59.9	58.5	57.3	68.2
letter	96.7	95.1	89.2	88.7	78.2	81.2	87.5	63.3	65.1	59.7
Lymphography	99.1	99.0	99.1	98.6	99.1	99.0	99.1	99.3	95.4	99.5
magic.gamma	85.8	86.6	85.3	81.1	80.9	81.3	77.0	77.5	72.0	73.4
mammography	87.8	87.6	88.4	85.9	86.4	88.7	88.9	87.9	84.9	85.9
mnist	93.5	94.7	93.5	95.5	87.9	89.5	87.6	85.9	91.2	66.7
musk	100.0	100.0	100.0	100.0	100.0	100.0	100.0	91.7	84.4	100.0
optdigits	99.2	98.2	99.5	99.7	90.9	85.2	88.2	83.9	64.2	87.5
PageBlocks	98.2	97.6	95.3	96.2	92.1	95.6	93.9	92.5	89.8	80.9
pendigits	99.4	99.4	99.9	99.4	97.8	97.8	98.8	96.5	90.8	93.8
Pima	72.7	72.4	75.1	71.7	72.2	69.3	61.0	72.3	71.1	72.1
satellite	84.7	85.8	87.7	85.0	85.0	80.0	82.5	81.3	73.8	86.6
satimage-2	99.8	99.9	99.9	99.7	99.8	99.9	99.9	99.5	92.7	98.2
shuttle	100.0	100.0	99.7	99.7	98.9	99.1	99.7	99.6	91.1	99.0
skin	97.7	96.3	98.7	77.9	92.1	92.3	55.2	88.6	51.7	77.4
smtp	74.3	76.2	88.5	85.5	82.9	80.7	69.9	85.8	71.4	84.9
SpamBase	83.0	78.4	77.0	82.2	73.0	73.7	83.3	86.4	88.7	77.0
speech	50.6	48.6	52.0	51.1	47.6	52.6	51.6	47.9	49.5	47.7
Stamps	96.1	93.8	93.8	92.3	93.6	92.5	88.4	94.4	91.5	92.7
thyroid	97.6	98.0	97.9	98.3	96.6	97.8	97.2	98.8	94.6	98.4
vertebral	47.0	45.3	43.7	46.3	50.9	46.1	50.3	42.0	52.3	38.1
vowels	98.8	97.1	98.1	97.1	93.8	93.5	94.4	78.8	76.9	70.2
Waveform	73.6	78.0	77.7	77.4	74.3	67.7	64.4	71.9	80.9	70.0
WBC	99.2	99.2	99.4	96.9	99.4	97.5	99.3	99.4	98.0	99.0
WDBC	99.7	99.6	99.5	99.8	99.4	98.8	99.5	99.5	96.2	99.0
Wilt	90.5	91.7	54.0	61.4	37.1	32.9	37.5	49.9	54.6	43.3
wine	94.4	93.7	96.7	96.3	96.3	96.1	95.5	92.3	83.1	90.4
WPBC	53.3	53.8	57.3	56.8	52.2	54.9	48.6	53.7	52.9	57.9
yeast	46.9	46.2	42.9	46.3	45.6	42.9	52.5	39.8	44.4	40.9
<i>Mean</i>	85.9	85.2	85.1	82.9	81.7	81.5	80.8	79.6	78.8	77.4

Table 7: Per-dataset AUROC ($\times 100$, five-seed means), clean regime: the remaining methods, continuing Table 6 over the same 47 datasets. [†]uTabPFN covers 44 of 47 datasets.

Dataset	PCA	OCSVM	DeepSVDD ^d	COPOD	ECOD	uTabPFN ^{††}	LODA	SOD	COF
ALOI	55.6	54.8	55.4	51.7	53.3	63.3	51.5	59.0	63.2
annthyroid	80.9	66.1	84.9	77.4	78.7	93.7	58.7	71.9	64.1
backdoor	69.4	84.7	58.7	78.4	84.2	36.5	66.9	69.5	73.7
breastw	99.1	99.6	99.0	99.7	99.5	6.6	99.5	86.9	56.8
campaign	77.0	69.1	72.9	78.1	77.0	53.1	55.7	65.5	52.5
cardio	96.4	97.4	93.6	92.0	93.6	87.5	91.3	49.0	53.4
Cardiotocography	82.8	85.2	79.5	66.2	78.5	82.4	80.1	35.5	41.6
celeba	80.6	72.1	74.8	76.0	76.6	44.1	62.5	48.3	39.5
census	71.0	56.1	69.5	66.5	65.6	–	44.3	58.2	41.0
cover	94.3	95.0	86.3	88.4	92.0	92.4	92.3	57.9	50.7
donors	88.0	88.3	84.1	81.4	89.1	68.0	46.8	56.1	69.6
fault	55.0	55.9	56.7	45.4	47.7	71.4	51.6	50.7	50.9
fraud	90.1	89.9	88.6	88.9	88.9	92.4	89.7	91.8	91.5
glass	73.9	47.4	64.5	78.1	72.2	88.7	71.4	76.5	82.2
Hepatitis	84.8	81.5	76.5	82.4	81.7	76.5	63.6	46.6	46.6
http	99.9	99.5	99.8	98.9	97.8	99.9	34.6	47.3	2.1
InternetAds	80.1	71.3	77.7	68.3	68.3	–	55.6	35.6	46.5
Ionosphere	89.9	92.9	84.5	80.0	74.1	95.7	88.6	73.2	75.4
landsat	39.4	37.2	41.6	41.1	35.8	75.2	41.0	45.3	50.7
letter	53.6	51.8	50.0	56.1	57.2	18.4	54.3	78.5	81.7
Lymphography	99.1	99.0	98.1	99.5	99.3	9.4	37.2	97.2	98.8
magic.gamma	70.6	71.9	67.6	68.6	64.6	88.1	73.0	66.9	59.7
mammography	90.5	89.9	85.4	90.5	90.6	87.8	89.5	78.2	67.8
mnist	90.5	88.6	80.6	77.4	74.8	78.5	82.0	61.6	57.7
musk	100.0	100.0	98.5	94.8	95.8	97.7	99.9	54.4	52.7
optdigits	58.0	64.5	54.7	68.4	60.6	32.1	66.7	44.4	56.9
PageBlocks	93.2	92.0	92.8	87.5	91.5	88.4	89.0	56.5	62.3
pendigits	94.2	96.4	86.9	90.5	92.7	99.7	93.4	51.3	51.5
Pima	69.9	67.7	67.2	64.7	60.1	63.9	67.6	50.1	54.9
satellite	67.0	65.6	68.0	63.2	58.4	86.9	68.7	47.5	51.5
satimage-2	97.8	98.2	97.2	97.5	96.5	99.7	98.4	74.3	55.9
shuttle	99.3	98.6	99.2	99.5	99.3	12.8	92.7	42.6	55.6
skin	26.1	75.3	60.7	45.9	48.2	99.2	74.5	42.4	40.7
smtp	70.5	66.2	68.1	87.4	81.1	83.8	66.5	60.6	77.9
SpamBase	80.4	72.4	79.9	67.7	64.2	78.2	59.8	32.1	18.9
speech	47.5	47.0	52.1	49.7	47.5	–	48.1	43.9	37.3
Stamps	93.9	91.3	94.3	93.2	88.2	94.9	92.3	57.2	36.8
thyroid	98.0	93.6	98.5	93.7	97.6	98.5	91.4	87.3	51.8
vertebral	44.1	42.7	42.1	35.6	42.3	48.0	36.3	38.0	29.9
vowels	61.7	62.3	57.3	49.7	59.3	99.9	71.8	73.8	79.6
Waveform	65.6	56.4	58.8	73.3	60.4	78.4	61.7	56.4	61.0
WBC	99.2	99.4	98.4	99.3	99.3	7.2	97.9	91.1	91.9
WDBC	99.5	99.7	98.9	99.5	97.1	99.7	99.1	87.2	92.8
Wilt	39.2	35.8	35.3	34.7	39.7	90.7	33.7	47.6	47.7
wine	92.3	93.7	83.1	85.6	73.2	97.2	87.7	30.8	16.3
WPBC	51.9	51.8	46.4	53.2	50.9	53.9	54.8	48.2	44.4
yeast	41.7	39.5	41.5	36.4	42.8	46.1	46.5	36.2	33.0
Mean	76.7	75.6	74.7	74.5	74.2	72.0	69.8	58.7	55.7

Table 8: AUPRC ($\times 100$) and mean per-dataset AUPRC rank over the 47 datasets, both regimes. Ranks cover the methods with values on all 47 datasets in that regime. A value appears only where the stored score vectors reproduce that method’s official AUROC (tolerance 0.25). ^a TCCM’s contaminated-regime score vectors were not retained, so its AUPRC there is withheld (Appendix D.2). [†] uTabPFN covers 44 datasets and is excluded from ranks.

Method	Contam.		Clean	
	AUPRC	rank	AUPRC	rank
FOCUS (ours)	41.7	5.6	78.3	4.5
ZEN (ours)	41.1	6.3	76.9	5.3
IForest	39.1	7.7	62.3	9.1
PCA	38.2	8.6	64.6	9.5
HBOS	37.1	7.9	61.7	9.8
ECOD	36.4	8.5	60.2	11.7
CBLOF	35.8	8.4	68.1	7.6
COPOD	35.7	8.7	58.3	12.0
DeepSVDD ^d	35.4	10.1	62.0	11.6
raw k NN	35.3	7.9	76.2	4.3
OCSVM	34.5	10.1	63.8	10.1
DTE ^d	32.9	9.6	70.1	7.9
TabPFN-PL [†]	32.7	9.6	52.6	11.7
LODA	30.3	10.9	56.3	12.6
uTabPFN ^{††}	29.8	–	63.4	–
SOD	28.9	10.6	44.2	15.4
COF	26.9	11.4	43.0	15.4
LOF	25.4	11.2	74.1	5.9
TCCM ^{da}	–	–	70.9	6.6

Table 9: Seed variability of every method in Tables 4–7: the 47-dataset mean AUROC ($\times 100$), the standard deviation of that mean over the five seeds, and the median over datasets of the per-dataset standard deviation over seeds; both regimes. Markers as in Table 1. ^a TCCM’s contaminated-regime per-seed values were not retained (Appendix D.2); the mean is Table 1’s. [†] uTabPFN covers 44 of 47 datasets.

Method	Contaminated			Clean		
	mean	sd of mean	median sd	mean	sd of mean	median sd
FOCUS (ours)	80.0	0.96	2.9	85.9	0.51	1.2
ZEN (ours)	79.7	0.84	2.8	85.2	0.38	1.5
IForest	76.4	0.44	2.3	79.6	0.46	1.8
TCCM ^d	72.2 ^a	–	–	80.8	0.69	2.0
TabPFN-PL [†]	74.6	0.45	1.8	78.8	0.17	1.3
COPOD	74.4	0.35	1.7	74.5	0.47	1.0
HBOS	74.4	0.23	2.2	77.4	0.57	1.3
DTE ^d	74.3	0.73	3.4	81.5	0.47	1.6
raw k NN	73.9	0.16	2.1	85.1	0.56	0.7
ECOD	73.9	0.52	2.0	74.2	0.36	0.8
CBLOF	73.7	0.54	2.9	81.8	0.83	1.6
PCA	72.5	0.95	1.9	76.7	0.45	1.0
DeepSVDD ^d	70.9	2.31	5.5	74.7	1.12	4.1
OCSVM	70.1	1.28	2.0	75.6	0.39	0.7
SOD	69.1	0.80	3.4	58.7	0.96	2.0
uTabPFN ^{††}	68.3	1.06	3.2	72.0	0.50	1.7
COF	66.4	0.94	3.3	55.7	0.74	2.0
LODA	66.0	1.20	6.8	69.8	1.04	4.6
LOF	65.4	0.65	3.2	82.9	0.58	1.1

C.3 REGRET PROFILES AND THE REMAINING GAP

Figure 7 shows how far each method trails the per-dataset best across the 47 datasets. On each dataset, a method’s regret is the AUROC of the best compared method on that dataset minus its own. The profile plots, for every tolerance τ , the fraction of datasets on which the method’s regret

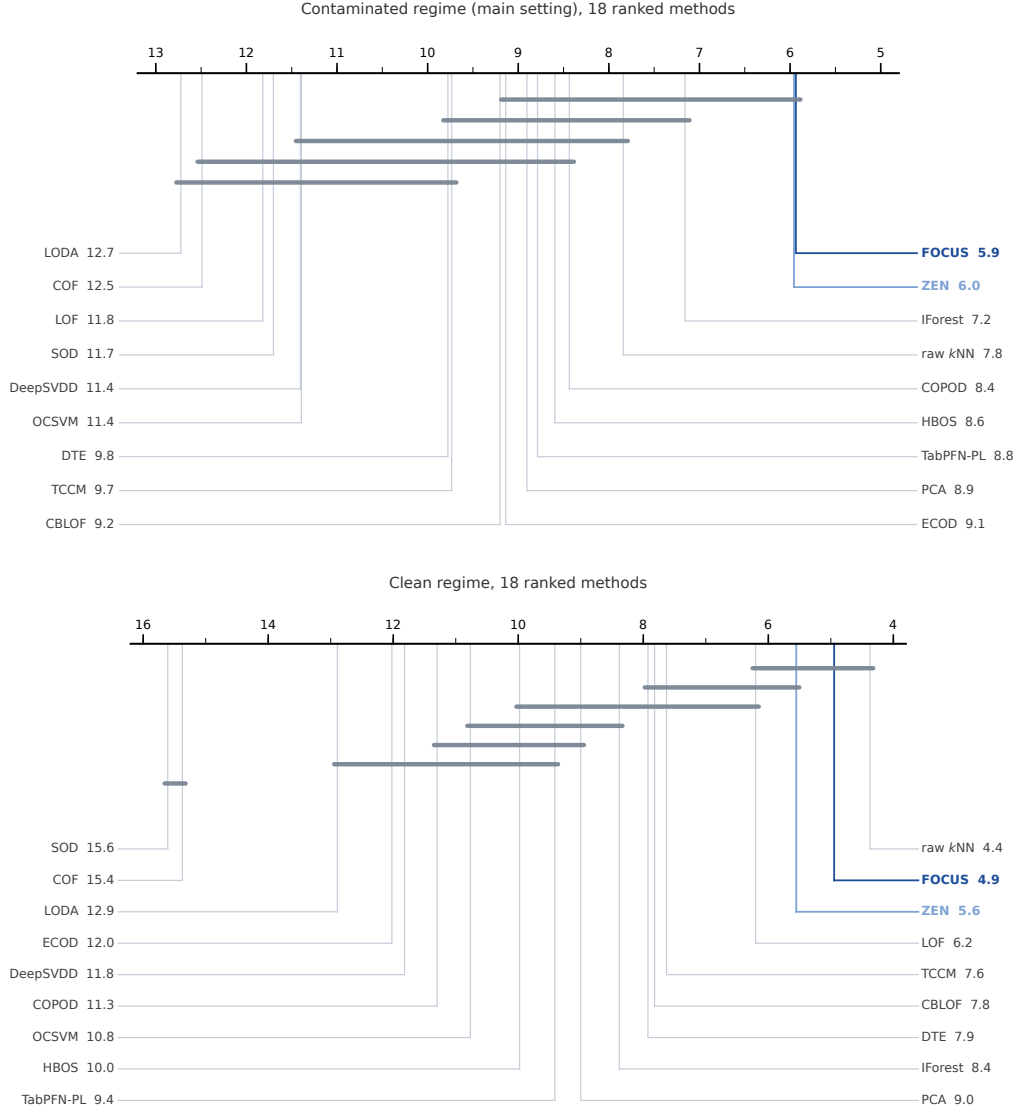


Figure 5: Critical-difference diagrams over the 47 datasets, best mean rank on the right; bars join methods whose pairwise Wilcoxon differences are not significant after Holm correction over all pairs (153 in each regime; we use Holm-corrected pairwise tests rather than the mean-rank post-hoc test, following Benavoli et al., 2016). Top: contaminated regime, 18 ranked methods. Bottom: clean regime, 18 ranked methods. uTabPFN is excluded from the ranks.

is at most τ ; a curve that rises faster belongs to a method that is rarely far from the best. Compared with an oracle that picks the best method for each dataset, FOCUS trails by 4.3 AUROC points on average and ZEN by 4.6, the two smallest gaps in the comparison. The largest gaps appear in two situations. On shuttle, Cardiotocography, vertebral and pendigits, four datasets with at most 21 features, Isolation Forest is well ahead of our detectors. On Wilt, the point above the diagonal in Figure 4b, the reference anomalies do not look isolated to the estimate, so the cleaning misfires and costs 6 AUROC points; over all 47 datasets the cleaning helps by more than one AUROC point on 27 and costs more than one point on 10. On Speech no method in the comparison, ours included, reaches 55 AUROC in either regime; vertebral and yeast are the only other such datasets. Speech’s features are nearly Gaussian, so the challenge there is the representation itself, and semantic context in the spirit of ReTabAD (Yoon et al., 2026) is a promising complement.

FOCUS 80.0	+3.62 28 / 0 / 19 p=0.010	+5.41 32 / 0 / 15 p=0.017	+5.57 29 / 0 / 18 p=0.008	+5.58 32 / 0 / 15 p=0.002	+5.72 34 / 1 / 12 p<0.001	+6.10 26 / 0 / 21 p=0.019	+6.11 33 / 1 / 13 p=0.002	+6.29 32 / 0 / 15 p<0.001	+7.48 30 / 1 / 16 p=0.002
	IForest 76.4	TabPFN-PL 74.6	COPOD 74.5	HBOS 74.4	DTE 74.3	raw kNN 73.9	ECOD 73.9	CBLOF 73.7	PCA 72.5
ZEN 79.7	+3.28 28 / 0 / 19 p=0.019	+5.07 29 / 0 / 18 p=0.035	+5.23 29 / 0 / 18 p=0.009	+5.23 30 / 1 / 16 p=0.003	+5.38 34 / 1 / 12 p<0.001	+5.76 28 / 0 / 19 p=0.024	+5.77 33 / 1 / 13 p=0.004	+5.94 34 / 1 / 12 p<0.001	+7.14 29 / 3 / 15 p=0.004
FOCUS 80.0	+7.79 36 / 0 / 11 p<0.001	+9.09 37 / 0 / 10 p<0.001	+9.88 36 / 0 / 11 p<0.001	+10.89 39 / 0 / 8 p<0.001	+13.11 33 / 0 / 11 p<0.001	+13.65 40 / 0 / 7 p<0.001	+14.06 38 / 0 / 9 p<0.001	+14.63 37 / 1 / 9 p<0.001	
	TCCM 72.2	DeepSVDD 70.9	OCSVM 70.1	SOD 69.1	uTabPFN [†] 68.3	COF 66.4	LODA 66.0	LOF 65.4	
ZEN 79.7	+7.44 36 / 0 / 11 p<0.001	+8.75 36 / 0 / 11 p<0.001	+9.53 36 / 0 / 11 p<0.001	+10.55 41 / 0 / 6 p<0.001	+12.77 34 / 0 / 10 p<0.001	+13.31 42 / 0 / 5 p<0.001	+13.71 37 / 0 / 10 p<0.001	+14.29 39 / 1 / 7 p<0.001	

Figure 6: Multi-comparison matrix of our methods against the seventeen baselines, contaminated regime, split into two column halves, in the format of Ismail-Fawaz et al. (2023) as adopted by recent large-scale benchmark studies (Middlehurst et al., 2024). Each cell gives the mean per-dataset AUROC difference (row minus column), the win/tie/loss count for the row method, and the Wilcoxon p-value as computed, before any correction, all in bold when $p < 0.05$; cell color encodes the mean difference (blue: row ahead; orange: column ahead). Rows and columns are ordered by mean AUROC. [†]uTabPFN covers 44 of 47 datasets.

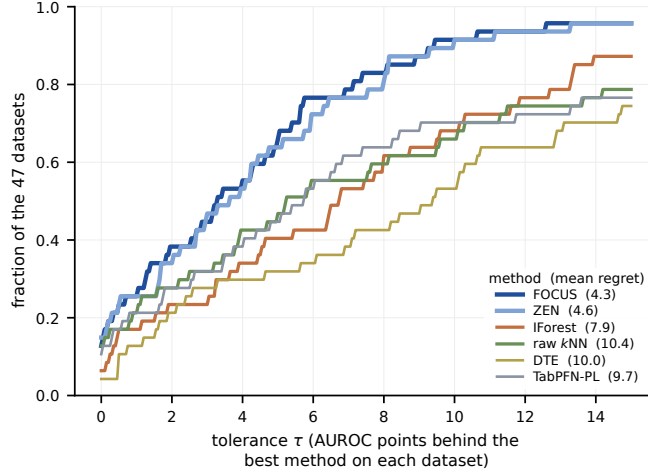


Figure 7: Regret profiles, contaminated regime, adapted from the performance profiles of Dolan & Moré (2002) with their runtime ratio replaced by the AUROC gap: the fraction of the 47 datasets on which a method is within τ AUROC points of that dataset’s best method. Mean regret in the legend; higher and further left is better. The per-dataset best is an oracle that changes identity across datasets; every method trails it somewhere, ours least. Shown: our methods and the strongest baselines.

C.4 ABLATIONS: EVERY STEP OF ZEN AND FOCUS

Table 10 lists the ladder of Figure 4a for both regimes and both models, recomputed on CPU from the official run’s saved embeddings with the official constants; the last step reproduces the ZEN and FOCUS means of the main runs to within 0.05 AUROC (half-precision storage). Each step is compared with the previous one by a Wilcoxon signed-rank test over the 47 dataset means, and Figure 8 plots the difference between the two models at each step, the fine-tune’s gain, in both regimes. The extra line of each contaminated block cleans every one of the 24 representations with its own weights instead of the pooled ones; Appendix C.6 varies every constant of the readout on

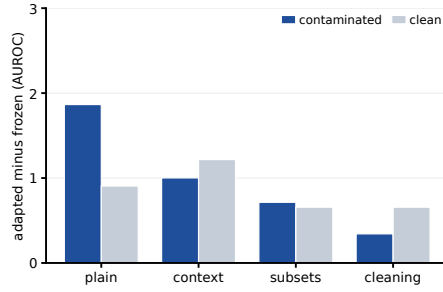


Figure 8: The fine-tune’s gain at each step of the ladder: the adapted model minus the frozen model, mean AUROC over the 47 datasets, both regimes.

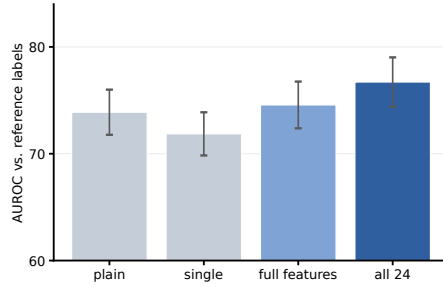


Figure 9: The isolation estimate against the reference labels, contaminated regime: AUROC of the estimate as an anomaly score for the reference samples, computed from one plain embedding, from one representation (one feature subset, one block), from the full set of features pooled over its four blocks, and from all 24 representations pooled as in ZEN; bars are standard errors.

the same runs. The isolation estimate of Sec. 2.2 ranks the reference samples of each contaminated run (Figure 9); against the reference labels, one plain embedding scores 73.9 AUROC, the average single representation (one feature subset, one block) 71.9, the full set of features pooled over its four blocks 74.6, the twenty random-subset representations pooled 76.2, and all 24 pooled 76.7. The pooled estimate beats the single representations on 43 of the 47 datasets ($p < 10^{-10}$). Among the 5% of reference samples the pooled estimate distrusts most, 31% are anomalies, against 12% in the reference set as a whole, and the trust weights built from it average 0.70 on a reference anomaly and 1.16 on a normal sample. The labels enter this diagnostic only; no method sees them.

What the fine-tune adds. The more of ZEN’s steps are in place, the less the fine-tune adds. Figure 8 reports FOCUS’s gain over the frozen model at each step of the ladder: about 1.9 AUROC points on the plain readout under contamination, 0.3 after the soft cleaning. One possible reading, which we have not tested directly, is that the fine-tune and the soft cleaning repair the same defect, a reference set polluted by anomalies. On clean data, where there is nothing to clean, the fine-tune keeps a significant gain of 0.6 to 1.2 AUROC points at every step.

C.5 ABLATION: THE READOUT WINDOW

The readout takes the embeddings of the last four of TabPFN-3’s 24 transformer blocks (Sec. 2.2). Table 11 and Figure 10 vary this choice on the same datasets, seeds and splits as the main results, in both regimes, for the frozen model (ZEN) and the adapted one (FOCUS): the last n blocks for n from 1 to 24, three four-block windows deeper in the network, and, in the figure, every single block. Every other constant is the paper’s, and the adapted model is the saved fine-tuned model of each run of the main evaluation, so no fine-tune is rerun. The four-block window reproduces the official means to within 0.05 AUROC in every case. Under contamination the last one to four blocks are equivalent, within 0.3 AUROC and never significantly different, and every wider or deeper window is worse; reading all 24 blocks costs 2.7 points for ZEN and 2.9 for FOCUS, both significant, and single blocks improve steadily with depth from block 8 on. On clean data the picture is flatter: the frozen readout gains 0.1 to 0.2 points from wider windows (last 6 to 12 blocks, p between 0.02 and 0.04) and loses 0.2 to 0.5 from narrower ones, while the adapted model is within 0.05 AUROC of its

Table 10: The ablation ladder of Figure 4a in numbers: mean AUROC over the 47 datasets at each step, the change from the previous step with its Wilcoxon p -value, and the number of datasets that improve. The last step of each block is the reported method (Table 1’s clean FOCUS, 85.9, recomputes here as 85.8, within the stated tolerance); in the clean regime the soft cleaning is switched off ($\lambda = 0$), so its row changes nothing.

Step	AUROC	vs previous	p	wins
<i>Contaminated regime, frozen model (ZEN)</i>				
plain readout	74.1	—	—	—
+ augmented context	75.0	+0.9	0.130	27/47
+ feature subsets	76.9	+1.9	<0.001	38/47
+ soft cleaning	79.7	+2.8	<0.001	36/47
soft cleaning per representation, not pooled	78.6	-1.1	<0.001	10/47
<i>Contaminated regime, adapted model (FOCUS)</i>				
plain readout	75.9	—	—	—
+ augmented context	76.0	+0.1	0.216	25/47
+ feature subsets	77.6	+1.6	<0.001	34/47
+ soft cleaning	80.0	+2.5	<0.001	35/47
soft cleaning per representation, not pooled	79.2	-0.9	<0.001	12/47
<i>Clean regime, frozen model (ZEN)</i>				
plain readout	84.2	—	—	—
+ augmented context	84.3	+0.1	0.698	26/47
+ feature subsets	85.2	+0.9	0.003	31/47
+ soft cleaning (off in this regime)	85.2	—	—	—
<i>Clean regime, adapted model (FOCUS)</i>				
plain readout	85.1	—	—	—
+ augmented context	85.5	+0.4	0.203	29/47
+ feature subsets	85.8	+0.3	0.011	33/47
+ soft cleaning (off in this regime)	85.8	—	—	—

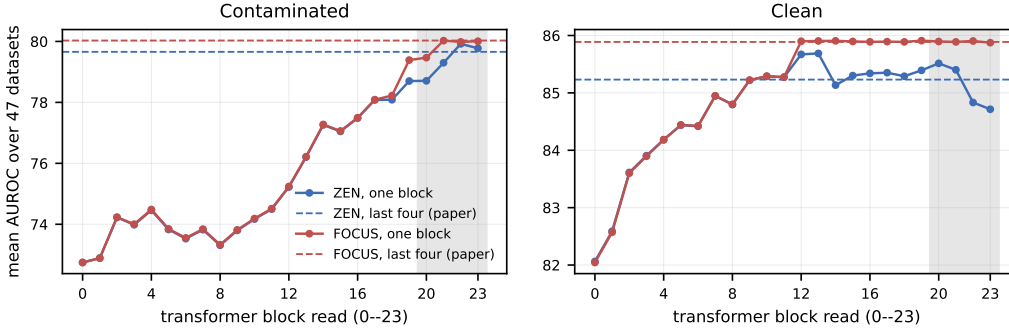


Figure 10: Mean AUROC over the 47 datasets, five seeds, when the readout uses one transformer block at a time (dots), for the frozen model (ZEN) and the adapted one (FOCUS); dashed lines: the paper’s four-block window, shaded.

reported value for every window from one to twelve blocks. The last-four window is therefore not a tuned optimum: under contamination it ties the best windows, and on clean data the best window is 0.3 points away.

C.6 SENSITIVITY OF THE READOUT

Table 12 varies each constant of the readout while every other constant stays at the paper’s value, on the saved embeddings of the main runs in both regimes, five seeds, for the frozen and the adapted model: the neighbor count k , the soft-cleaning strength λ , the mixing weight α , where the score is α times the random-subset average plus $1 - \alpha$ times the full-feature score, so that $\alpha = 0.5$ is Eq. 4 up to a constant factor, and the number B of random feature subsets. The paper’s setting reproduces the official means to within 0.05 AUROC. Under contamination $k = 50$ sits on a plateau: $k = 100$ and 200 are within 0.2 points (not significant) and smaller k loses up to 2.8. On clean data every k from 5 to 200 stays within 0.8 points of the paper’s value, the smaller values marginally above it;

Table 11: The readout window: mean AUROC over the 47 datasets, five seeds, for the frozen model (ZEN) and the adapted one (FOCUS) in both regimes, with the difference from the paper’s window and its Wilcoxon p -value. Blocks are numbered 0–23; the paper reads blocks 20–23.

Readout window	Contaminated		Clean	
	ZEN	FOCUS	ZEN	FOCUS
last block	79.8 (+0.1, 0.65)	80.0 (± 0.0 , 0.60)	84.7 (-0.5, <0.001)	85.9 (± 0.0 , 0.06)
last 2 blocks	79.9 (+0.3, 0.91)	80.0 (± 0.0 , 0.50)	84.8 (-0.4, <0.001)	85.9 (± 0.0 , 0.06)
last 3 blocks	79.8 (+0.2, 0.54)	80.0 (± 0.0 , 0.49)	85.0 (-0.2, <0.001)	85.9 (± 0.0 , 0.13)
last 4 blocks (paper)	79.7	80.0	85.2	85.9
last 6 blocks	79.3 (-0.4, 0.22)	79.7 (-0.3, 0.37)	85.3 (+0.1, 0.04)	85.9 (± 0.0 , 0.48)
last 8 blocks	79.0 (-0.6, 0.19)	79.4 (-0.7, 0.36)	85.3 (+0.1, 0.03)	85.9 (± 0.0 , 0.03)
last 12 blocks	78.5 (-1.2, 0.18)	78.7 (-1.4, 0.33)	85.5 (+0.2, 0.02)	85.9 (± 0.0 , 0.54)
all 24 blocks	77.0 (-2.7, 0.04)	77.1 (-2.9, 0.02)	85.1 (-0.1, 0.46)	85.5 (-0.4, 0.09)
blocks 16–19	78.2 (-1.5, 0.07)	78.4 (-1.6, 0.10)	85.4 (+0.1, 0.07)	85.9 (± 0.0 , 0.21)
blocks 12–15	76.6 (-3.1, 0.02)	76.6 (-3.4, 0.02)	85.5 (+0.3, 0.03)	85.9 (± 0.0 , 0.45)
blocks 8–11	74.0 (-5.7, 0.01)	74.0 (-6.1, 0.004)	85.2 (± 0.0 , 0.54)	85.2 (-0.7, 0.06)

Table 12: Sensitivity of the readout: mean AUROC over the 47 datasets, five seeds, for the frozen model (ZEN) and the adapted one (FOCUS) in both regimes, one constant varied at a time with the difference from the paper’s value and its Wilcoxon p -value. With B random subsets the trust weight pools the $(B+1) \times 4$ representations, as ZEN does with six.

	Contaminated		Clean	
	ZEN	FOCUS	ZEN	FOCUS
<i>Neighbor count k</i>				
5	76.9 (-2.8, <0.001)	77.9 (-2.2, 0.002)	86.0 (+0.8, 0.03)	86.2 (+0.3, 0.35)
10	78.0 (-1.7, 0.001)	78.8 (-1.2, 0.02)	85.8 (+0.6, 0.01)	86.2 (+0.3, 0.10)
20	78.8 (-0.9, 0.004)	79.5 (-0.5, 0.06)	85.5 (+0.3, 0.007)	86.1 (+0.2, 0.05)
50 (paper)	79.7	80.0	85.2	85.8
100	79.9 (+0.2, 0.45)	80.1 (+0.1, 0.83)	85.0 (-0.2, 0.004)	85.8 (-0.1, 0.11)
200	79.7 (+0.1, 0.96)	79.6 (-0.4, 0.15)	84.8 (-0.4, 0.009)	85.8 (-0.1, 0.12)
<i>Soft-cleaning strength λ</i>				
0 (paper, clean)	76.9 (-2.8, <0.001)	77.6 (-2.5, <0.001)	85.2	85.8
0.25	79.3 (-0.4, 0.001)	79.7 (-0.3, 0.008)	84.8 (-0.4, 0.10)	84.6 (-1.2, 0.02)
0.5 (paper, contam.)	79.7	80.0	84.5 (-0.7, 0.04)	84.1 (-1.7, 0.005)
1	79.7 (± 0.0 , 0.76)	80.1 (± 0.0 , 0.81)	84.0 (-1.1, 0.01)	83.7 (-2.1, 0.001)
2	79.3 (-0.4, 0.42)	79.4 (-0.6, 0.58)	83.4 (-1.8, <0.001)	83.4 (-2.5, <0.001)
<i>Weight of the random subsets α</i>				
0 (full set only)	79.1 (-0.6, 0.002)	79.5 (-0.6, 0.009)	84.3 (-0.9, 0.003)	85.5 (-0.3, 0.01)
0.25	79.5 (-0.2, 0.02)	79.9 (-0.2, 0.10)	84.8 (-0.3, 0.01)	85.8 (± 0.0 , 0.19)
0.5 (paper)	79.7	80.0	85.2	85.8
0.75	79.6 (-0.1, 0.81)	80.0 (-0.1, 0.97)	85.0 (-0.2, 0.42)	85.5 (-0.3, 0.04)
1 (random subsets only)	79.2 (-0.5, 0.05)	79.5 (-0.6, 0.02)	84.0 (-1.2, 0.005)	84.5 (-1.4, <0.001)
<i>Number of random feature subsets B</i>				
0	77.5 (-2.2, <0.001)	77.9 (-2.1, <0.001)	84.3 (-0.9, 0.003)	85.5 (-0.3, 0.01)
1	77.9 (-1.8, <0.001)	78.4 (-1.6, <0.001)	84.1 (-1.1, <0.001)	85.1 (-0.7, <0.001)
2	78.6 (-1.1, <0.001)	79.0 (-1.0, <0.001)	84.8 (-0.4, 0.005)	85.6 (-0.3, 0.004)
3	79.2 (-0.4, 0.04)	79.6 (-0.4, 0.04)	85.0 (-0.2, 0.02)	85.7 (-0.1, 0.02)
4	79.5 (-0.2, 0.59)	79.8 (-0.2, 0.34)	85.1 (-0.1, 0.14)	85.8 (-0.1, 0.11)
5 (paper)	79.7	80.0	85.2	85.8

one k serves both regimes. Every λ from 0.25 to 2 improves on no cleaning under contamination and stays above Isolation Forest, by 2.9 to 3.3 points for ZEN, and 0.5 and 1 give the same mean, so the paper’s value sits on a plateau rather than a peak; on clean data every $\lambda > 0$ costs 0.4 to 2.5 points, which is why the cleaning is switched off there. The equal mixing weight is best in both regimes, but α between 0.25 and 0.75 stays within 0.3 points, while the full set of features alone or the random subsets alone cost 0.3 to 1.4. Under contamination each added random subset helps, significantly up to the fourth; the fifth adds 0.2 points (not significant), and three random subsets already place ZEN above every baseline in Table 1. On clean data the first random subset alone is no better than the full set of features alone, and from the second subset on the average helps.

C.7 ABLATIONS OF THE FINE-TUNE

Table 13 varies the constants of the fine-tune: the number K of trainable blocks, the trust weights of Eq. 6, the number of epochs, the checkpoint rule and the learning rate. These arms rerun the

Table 13: Ablations of the fine-tune, seed 0, 47 datasets, mixed GPU types: mean AUROC of FOCUS with the difference from the paper’s setting rerun under the same conditions and its Wilcoxon p -value. Every arm keeps the paper’s other constants; the last row is the frozen model on the same runs.

Fine-tuning variant	Contaminated	Clean
<i>Trainable blocks K (paper: 6 contaminated, 12 clean)</i>		
$K = 3$	80.5 (-0.5, 0.91)	85.5 (-0.5, 0.14)
$K = 6$	81.0 (paper)	85.0 (-1.0, 0.04)
$K = 12$	78.3 (-2.6, 0.004)	86.0 (paper)
$K = 24$	77.8 (-3.2, 0.002)	83.1 (-3.0, <0.001)
<i>Other constants, at the paper’s K</i>		
trust weights off (plain compactness loss)	80.8 (-0.2, 0.06)	85.8 (-0.2, 0.02)
1 epoch	80.9 (± 0.0 , 0.60)	85.9 (-0.2, 0.37)
3 epochs (three checkpoints averaged)	81.1 (+0.1, 0.66)	85.8 (-0.2, 0.03)
last epoch kept instead of the average	80.9 (± 0.0 , 0.95)	86.0 (-0.1, 0.68)
learning rate $\times 0.1$	80.6 (-0.4, 0.45)	86.2 (+0.1, 0.05)
learning rate $\times 10$	79.2 (-1.8, 0.05)	66.1 (-19.9, <0.001)
ZEN on the same runs	80.0	85.1

Table 14: The context of each run, contaminated regime, seed 0, 47 datasets: mean AUROC of ZEN with the difference from the paper’s setting and its Wilcoxon p -value.

Context construction	ZEN, contaminated
no synthetic samples ($m = 0$)	79.4 (-0.4, 0.18)
$m = 100$	79.2 (-0.7, 0.06)
$m = \min(n, 500)$, 10 folds (paper)	79.9
$m = \min(n, 2000)$	80.6 (+0.8, 0.56)
5 folds	80.5 (+0.6, 0.18)
2 folds	80.4 (+0.5, 0.92)

fine-tune, so they are run at one seed (seed 0) on every dataset, on the mixed GPU types of our cluster’s spare nodes. The frozen readout itself differs between GPU types by up to a few AUROC points on the smallest datasets, so the paper’s own setting is rerun under the same conditions and every arm is compared with that rerun, paired over the 47 datasets; the rerun reproduces the official seed-0 means to within 0.05 AUROC in both regimes. Under contamination $K = 3$ and the paper’s $K = 6$ are equivalent (-0.5 , $p = 0.91$) and deeper fine-tunes are worse, by 2.6 points at $K = 12$ and 3.2 at $K = 24$, both significant. On clean data the paper’s $K = 12$ is best, $K = 3$ is within 0.5 points (not significant), $K = 6$ costs 1.0 ($p = 0.04$) and $K = 24$ costs 3.0. A single $K = 3$ would therefore serve both regimes within 0.5 AUROC of the reported values. Switching the trust weights off costs 0.2 AUROC in both regimes ($p = 0.06$ and 0.02). Under contamination one or three epochs, or keeping the last epoch’s weights instead of the average, change the mean by at most 0.1 points (not significant); a learning rate ten times smaller costs 0.4 (not significant) and ten times larger costs 1.8 ($p = 0.05$). On clean data one or three epochs, or keeping the last epoch’s weights, change the mean by at most 0.2 points (ep3: $p = 0.03$); a learning rate ten times smaller changes it by +0.1 ($p = 0.05$) while a rate ten times larger destabilizes the fine-tune (-19.9 points, $p < 0.001$): lower rates than the paper’s are safe, higher ones are not.

C.8 ABLATION: THE CONTEXT OF EACH RUN

Table 14 varies the number m of synthetic context samples and the number of leave-one-fold-out folds of Sec. 2.2 for the frozen model under contamination, at seed 0 on every dataset (the arms re-embed the reference set, on mixed GPU types); the paper’s setting recomputed on the same runs is the reference, and it reproduces the official seed-0 mean to within 0.2 AUROC. Without synthetic samples ZEN loses 0.4 points and with 100 of them 0.7 ($p = 0.18$ and 0.06); with up to 2,000 it gains 0.8 (not significant; on datasets with at most 500 reference samples this arm equals the paper’s). The augmented context is thus a small, consistent component. Five or two folds instead of ten change the mean by +0.6 and +0.5 points, neither significant: the reference set can be embedded in two passes instead of ten at no measurable cost, the remedy Sec. 6 names.

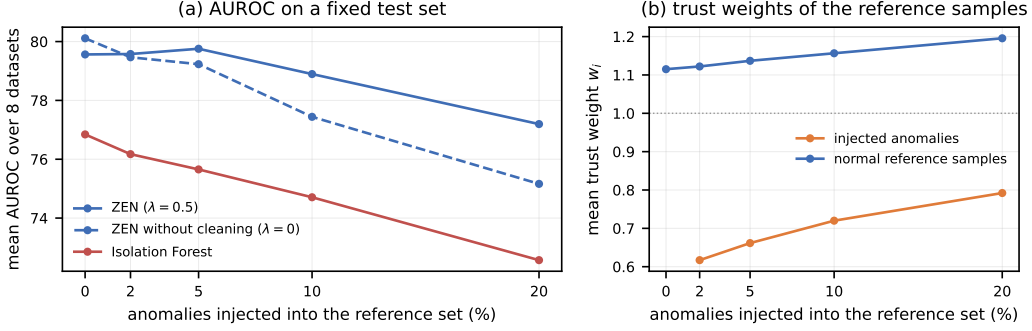


Figure 11: Injected contamination on eight datasets, five seeds. (a) Mean AUROC on a test set that is identical at every rate, as anomalies are injected into the clean reference set; ZEN with and without the soft cleaning, and Isolation Forest. (b) Mean trust weight of the injected anomalies and of the normal reference samples.

C.9 INJECTED CONTAMINATION

The contaminated regime of the main experiments carries each dataset’s natural anomaly rate. Figure 11 controls the rate instead, on the eight datasets whose anomaly rate is at least 30% (breastw, fault, Ionosphere, magic.gamma, Pima, satellite, SpamBase and yeast), the ones on which a 20% reference contamination is feasible while half of the anomalies stay in the test set. The reference set is the clean regime’s, normal samples only; half of the test anomalies, drawn once per seed, form an injection pool, and the first round ($\frac{r}{1-r} n$) of them are appended to the n reference normals for the rate r in $\{0, 2, 5, 10, 20\}\%$, so that the injected sets are nested and the test set, the clean test normals plus the other half of the anomalies, is identical at every rate. The feature scaler is fitted once on the clean reference set. We run ZEN with the contaminated readout ($\lambda = 0.5$), the same embeddings read without cleaning ($\lambda = 0$), and the strongest contaminated-regime baseline of Table 1, Isolation Forest, on five seeds; FOCUS is not rerun per rate. ZEN’s gain over its own uncleaned readout grows with the rate, from 0.1 points at 2% to 0.5, 1.5 and 2.0 at 5, 10 and 20% ($p = 0.08$ at 20% on eight datasets), and its margin over Isolation Forest grows from 2.7 to 4.6 points, significant from 5% on. On the clean reference set the cleaning costs 0.5 points (not significant), consistent with $\lambda = 0$ in the clean regime. Panel (b) shows the mechanism: the trust weight of the injected anomalies averages 0.62 to 0.79 against 1.12 to 1.20 for the normal reference samples, and the isolation estimate ranks the injected samples with 75 to 79 AUROC against the injection labels. SpamBase’s clean test set keeps four normal samples (Appendix A.1), so its curve is the noisiest of the eight.

C.10 ZEN ON OTHER BACKBONES

Table 15 and Figure 12 report the plain readout and ZEN on five backbones in both regimes: TabPFN-3, the paper’s backbone; TabPFN v2 (Hollmann et al., 2025); TabICL (Qu et al., 2025), in its TabICLv2 checkpoint (Qu et al., 2026); TabDPT (Ma et al., 2025); and Mitra (Zhang et al., 2025). Every constant of Sec. 2.2 and Appendix D.1 is unchanged, and every model runs frozen with the full reference set as its context, a single estimator and its test-time augmentations switched off. The one rule applied per backbone is where the embeddings are taken: the last four blocks of the model’s in-context transformer, at the point where the model itself uses its last block. These are blocks 20–23 of 24 for TabPFN-3, 8–11 of 12 for TabPFN v2, TabICL and Mitra, and 28–31 of 32 for TabDPT. Each block’s output passes through the model’s own final normalization where the model has one; TabDPT’s running representation is taken directly, and Mitra’s at the label slot of each query sample. We check on every run that the captured last block reproduces the model’s own output, to within 0.02 in predicted probability. TabDPT accepts at most 128 features, so on the five datasets with more features each run uses a fixed random subset of 128 of them, the same device as the paper’s 200-feature cap. Every run used the same GPU model and the paper’s five seeds. The TabPFN-3 arm reproduces the paper’s ZEN on every dataset to within 0.5 AUROC, and raw-feature k NN, which needs no forward pass, reproduces the paper’s values exactly.

Under contamination ZEN’s steps help on every backbone, with the largest gain on the weakest one. In the clean regime the soft cleaning is switched off and only the feature subsets and the augmented

Table 15: ZEN on five backbones: mean AUROC of the plain readout and of ZEN over the 47 datasets, ZEN’s gain with its Wilcoxon p -value and the number of datasets that improve, and the number of datasets on which ZEN beats raw-feature k NN. Every constant is the paper’s.

Backbone	blocks	plain	ZEN	gain (p , wins)	ZEN > raw k NN
<i>Contaminated regime</i>					
TabPFN v2	12	67.2	75.4	+8.2 (<0.001, 43/47)	22/47
Mitra	12	70.9	76.3	+5.4 (<0.001, 37/47)	22/47
TabICL	12	71.7	78.9	+7.2 (<0.001, 39/47)	26/47
TabDPT	32	74.2	79.0	+4.9 (<0.001, 36/47)	29/47
TabPFN-3	24	74.2	79.7	+5.5 (<0.001, 36/47)	28/47
raw-feature k NN (no backbone)	–	73.9	–	–	–
<i>Clean regime</i>					
TabPFN v2	12	76.4	79.9	+3.4 (<0.001, 38/47)	11/47
Mitra	12	83.1	84.2	+1.1 (0.003, 33/47)	20/47
TabICL	12	84.5	84.9	+0.4 (0.045, 31/47)	24/47
TabDPT	32	84.6	85.2	+0.6 (0.356, 24/47)	22/47
TabPFN-3	24	84.1	85.2	+1.1 (0.010, 28/47)	21/47
raw-feature k NN (no backbone)	–	85.1	–	–	–

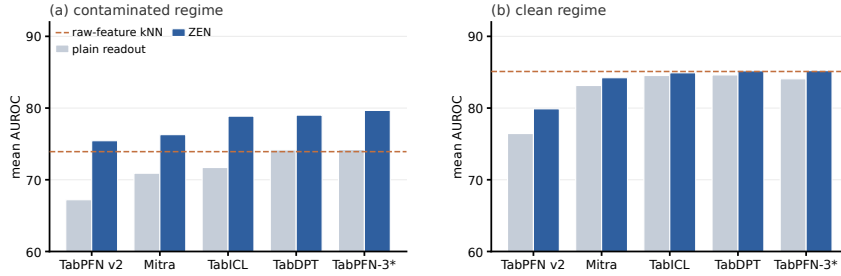


Figure 12: Plain readout and ZEN on five in-context backbones, both regimes; dashed line: raw-feature k NN; the asterisk marks the paper’s backbone, TabPFN-3. Figure 4c is panel (a) without the raw-feature k NN line.

context act; the gain is then smaller, significant for TabPFN-3, TabPFN v2, TabICL and Mitra, and not for TabDPT (0.6 AUROC points, $p = 0.36$). Between backbones, TabPFN-3 leads TabPFN v2 significantly in both regimes (by 4.2 and 5.3 AUROC points); its leads over TabICL (0.8, $p = 0.13$; 0.3, $p = 0.88$) and over TabDPT (0.7, $p = 0.26$; 0.0, $p = 0.78$) are not significant. Its lead over Mitra is significant under contamination (3.4, $p = 0.006$) and not on clean data (1.0, $p = 0.24$); Mitra in turn leads TabPFN v2 significantly on clean data (4.3, $p < 0.001$) but not under contamination (0.8, $p = 0.28$). On clean data TabPFN-3 (85.2) and TabDPT (85.2) reach raw-feature k NN (85.1), TabICL (84.9) and Mitra (84.2) stay just below it, and TabPFN v2 (79.9) well below.

FOCUS on TabPFN v2. The fine-tune was also run on TabPFN v2 under contamination, with the contaminated regime’s constants unchanged ($K = 6$ of its 12 blocks) and the paper’s five seeds. Three datasets (backdoor, census and speech) exceed the memory of our largest GPU on this arm and are left out, so the arm covers 44 datasets. There the fine-tune lifts the plain readout from 67.6 to 70.7 AUROC (+3.0, $p < 0.001$, 31 of 44 datasets), against +2.0 ($p = 0.002$) on TabPFN-3 in the same run, and FOCUS reaches 76.1 AUROC, above ZEN’s 75.9 on the same datasets: as on TabPFN-3, the fine-tune’s gain is large on the plain readout and small once ZEN’s steps are in place (Sec. 4). This arm ran on other GPU types than the runs of Table 15; the TabPFN-3 arm of the same run, on the paper’s GPU type, reproduces the paper’s FOCUS to within 0.05 AUROC.

D IMPLEMENTATION DETAILS

D.1 METHOD CONFIGURATION

Table 16 lists every constant of ZEN and FOCUS. Everything not marked per-regime is shared. Nothing in the table is ever chosen per dataset.

Table 16: Complete configuration of our methods. The upper block is shared by both regimes; the lower block lists the per-regime constants.

Backbone	TabPFN-3 (Grinsztajn et al., 2026), 24 transformer blocks (<code>tabpfn</code> package 8.1.0, default checkpoint)
Readout blocks	the last four of the 24
Feature subsets	five random subsets of size $\max(2, \text{round}(d/2))$ and the full set of features, drawn once per dataset and seed
Combination	mean of the random subsets + full set, equal weight (Eq. 4)
Synthetic context samples	$m = \min(n, 500)$ per feature subset, uniform on $[-1, 1]$, arbitrary labels, never scored
Neighbor count k	50
Soft-cleaning weight	$w_i = \exp(-\lambda \hat{a}_i)$, $\hat{a}_i$ pooled over the 24 (subset, block) representations (Sec. 2.2), clipped to $[10^{-3}, 10^3]$
Score standardization	per block; μ, σ from the reference set’s weighted leave-one-out k NN distances
Embedding extraction	leave-one-fold-out, 10 folds, 1 estimator
Context labels	fixed arbitrary (random 0/1, drawn per seed)
Trust weights (FOCUS)	same rule on the frozen final-block embedding of the reference set alone (all features, no synthetic samples), $k=20, \lambda=0.5$ in both regimes
Center c	trust-weighted mean of the frozen final-block embeddings
Loss (FOCUS)	trust-weighted squared distance to c (Eq. 6)
Epochs	2, no validation split, no early stopping
Kept weights	mean of the epoch-1 and epoch-2 parameters
Optimizer	AdamW, constant learning rate
Wide inputs	200-feature cap on three datasets (App. A.3)

	clean	contaminated
Soft-cleaning strength λ	0	0.5
Learning rate	10^{-3}	10^{-4}
Batch size	32	16
Trainable blocks K	12	6
Gradient clipping	none	0.01

The PFN pretraining objective. With D_{train} , x_{test} and q_θ as in Sec. 2.1, the parameters of a PFN minimize

$$\mathcal{L}_{\text{PFN}}(\theta) = \mathbb{E}_{D \sim p(D)} [-\log q_\theta(y_{\text{test}} | x_{\text{test}}, D_{\text{train}})], \quad (5)$$

where each synthetic dataset D drawn from the prior is split into a context D_{train} and held-out test pairs $(x_{\text{test}}, y_{\text{test}})$ (Hollmann et al., 2023).

Embedding extraction. TabPFN expects labeled context samples and anomaly detection provides none, so each run passes fixed arbitrary labels: a random 0/1 vector drawn once from the run’s seed. The labels carry no information about which samples are anomalous, and the readout never uses the classifier head. Reference-set embeddings are extracted leave-one-fold-out with ten folds, following Ye et al. (2025): each fold is embedded in the query role against the remaining folds, so training and test embeddings come from the same code path in the same role. When the reference set holds fewer than ten samples, the fold count drops to that number, never below two. Each extraction uses a single estimator for determinism.

Feature subsets and synthetic context samples. The B random feature subsets hold $\max(2, \text{round}(d/2))$ features each, drawn without replacement once per dataset and seed. Random feature subsets are the device behind random forests (Breiman, 2001) and feature-bagging outlier ensembles (Lazarevic & Kumar, 2005), and the motivation is the same here: runs that make different mistakes average out one another’s errors (Sec. 5). A run’s context is its feature subset of the reference set plus $m = \min(n, 500)$ synthetic samples drawn uniformly on $[-1, 1]$ over its features, with the same kind of arbitrary labels as the reference samples. The synthetic samples are never scored and never serve as neighbors; they only change what the model conditions on when it embeds the real samples, and they stay in every fold’s context. Figure 4a, Appendix C.4 and Appendix C.8 show their effect.

The FOCUS objective. Each reference sample x_i carries a trust weight w_i of the form of Sec. 2.2, computed on the frozen final-block embedding of the reference set alone, all features and no synthetic samples, with its own constants, $k=20$ and $\lambda=0.5$, the same in both regimes (the readout’s λ is 0 on clean data, the fine-tune’s is not). The same weights define the center as the weighted mean of the frozen embeddings and enter the loss that FOCUS minimizes over batches $\mathcal{B}$ of reference samples:

$$\mathcal{L}(\theta) = \frac{\sum_{x_i \in \mathcal{B}} w_i \|\psi_\theta(x_i | D_{\text{train}}) - c\|_2^2}{\sum_{x_i \in \mathcal{B}} w_i}, \quad c = \frac{\sum_i w_i \psi_\theta(x_i | D_{\text{train}})}{\sum_i w_i}. \quad (6)$$

Here ψ_θ is the extractor once its weights update and, without a block index, denotes the final-block embedding; the sums defining c run over the whole reference set. Weights and center are computed once from the frozen model and never updated: recomputing the weights as the model adapts would let it down-weight whatever it fails to contract, and a fixed center avoids the collapse failure mode of learned centers (Ruff et al., 2018). Normalizing by the weight sum keeps the loss scale, and with it the effective step size, independent of the trust strength. With all weights equal, Eq. 6 is the plain compactness loss; with the trust weights it is what we call *trust-weighted compactness fine-tuning*. FOCUS updates only the last K blocks of TabPFN and trains on the reference set alone, embedded leave-one-fold-out with the folds above, so it compacts a set that is also the model’s context; in the contaminated regime that set contains anomalies, and FOCUS is never told which samples they are.

Fine-tuning. FOCUS updates the last K of TabPFN’s 24 transformer blocks, $K=12$ in the clean regime and $K=6$ in the contaminated one, with the learning rate, batch size and gradient clipping of Table 16. Training runs for two epochs, and the kept model is the mean of the epoch-1 and epoch-2 parameters. Appendix C.7 ablates K , the trust weights, the epochs, the checkpoint rule and the learning rate.

How the constants were chosen. The training constants are simple, standard values, fixed once per regime before the reported five-seed runs; none is chosen per dataset. Raw-feature k NN uses $k=20$, inside the $10 < k < 50$ range recommended for neighbor-based detectors (Goldstein & Uchida, 2016). ZEN and FOCUS use $k=50$, one value for both regimes and every dataset.

D.2 BASELINE CONFIGURATIONS

ADBench benchmarks 14 unsupervised detectors, 13 of them through PyOD. DAGMM is the exception: ADBench ships its own implementation. We run 12 of the 13 PyOD detectors and leave DAGMM out rather than port a second implementation of one legacy baseline. PyOD’s k NN is replaced by raw-feature k NN, the same algorithm scored by the mean distance to its $k=20$ nearest neighbors (Appendix D.1) rather than the library default, the distance to the fifth neighbor. All PyOD detectors run at library defaults with contamination = 0.1, matching how ADBench itself instantiates them (Han et al., 2022), which likewise runs every detector at its default hyperparameters rather than tuning on a labeled hold-out set. Such tuning is not available to a label-free practitioner in any case, so no method is tuned per dataset (Sec. 3). The one deviation from the defaults is PCA’s component count. At the default the detector keeps every component, and on three (dataset, seed) pairs the training covariance is singular, which makes the resulting scores infinite. We therefore cap the count at 95% cumulative explained variance, a standard sklearn idiom, uniformly on every dataset.

The added baselines run as their authors specify. TCCM trains one model per dataset with its authors’ architecture and training protocol; its per-dataset epoch count and batch size come from the fixed table their code ships, one entry per dataset, unchanged across seeds. The per-seed score vectors of its contaminated-regime run were not retained, only the per-dataset AUROC means, so its contaminated AUPRC is withheld in Table 8. It reads the shared MinMax $[-1, 1]$ features like every other method rather than its own z-scoring default. DTE runs in its categorical variant at the authors’ recommended hyperparameters. The TabPFN pseudo-labeling control flags the 10% of training samples a PCA detector scores as most anomalous, fits TabPFN’s classifier on those pseudo-labels, and scores each test sample by its predicted anomaly probability. The official unsupervised TabPFN extension (uTabPFN) runs through its public API; its per-feature density construction is what makes it infeasible on the three widest datasets (App. A.3). The pretrained anomaly detectors of Sec. 5

Table 17: Wall-clock time over the 47 contaminated-regime datasets on one NVIDIA A100-SXM4-80GB. Baselines: fit plus scoring on the seed-0 splits, single run, indicative rather than seed-averaged. Ours: median per dataset and total per seed of each stage, over the 235 runs of the official run.

Method	median s / dataset	total s (47 datasets)
LODA	0.03	2.7
PCA	<0.01	3.1
COPOD	0.01	3.2
ECOD	0.01	3.3
raw k NN	0.05	3.6
HBOS	<0.01	4.5
CBLOF	0.03	4.5
LOF	0.08	6.5
IForest	0.16	7.8
OCSVM	0.70	70.6
SOD	1.37	95.2
DeepSVDD	6.64	325.5
COF	2.77	351.5
ZEN (ours): engines and reference embedding	16.6	850
ZEN (ours): six feature-subset embeddings and readout	57.3	2,904
FOCUS (ours): fine-tune (training)	77.3	7,340
FOCUS (ours): six feature-subset embeddings and readout	57.0	2,879

(TACTIC, FoMo-0D, OutFormer, OFA-TAD) are not run: this paper asks what a general-purpose tabular model gives anomaly detection, so every compared method starts without anomaly-specific pretraining.

D.3 RUNTIME

Table 17 reports wall-clock times on a single NVIDIA A100 GPU in the contaminated regime. The baseline rows come from one probe on the seed-0 splits; each is the detector’s fit plus scoring. The rows of our methods are stage medians over the 235 runs of the main evaluation (47 datasets, five seeds) on the same GPU model, with totals per seed. ZEN’s inference has two stages: building the TabPFN engines and embedding the reference set, then the six feature-subset embeddings and the k NN readout. FOCUS adds its one-time fine-tune, a training cost, and then runs the same inference on the adapted model. The probe does not cover TCCM, DTE, uTabPFN, or TabPFN-PL.

E NEGATIVE RESULTS

We record the design alternatives we tried and rejected. Each was evaluated under the paper’s protocol in the contaminated regime and improved on nothing.

- **LOF scoring instead of k NN distance.** Local-density normalization loses to plain k NN distance, on raw features and on our embeddings. Several datasets contain enough exact duplicate samples to break LOF’s reachability normalization, and a de-duplicated variant did not close the gap.
- **Hubness correction.** Rescaling the embedding distances by mutual proximity is worse than the plain distance.
- **Whitened and Mahalanobis-style k NN.** Not significantly better than plain k NN in either regime.
- **Distance to the center as the score.** The training objective itself, read directly as an anomaly score, is far worse than the k NN readout in both regimes.
- **Bagging the reference set.** Averaging the readout over random subsets of the reference set, instead of over feature subsets, brings nothing.
- **Other ways to pool the isolation estimate.** Weighting feature subsets by their agreement, pessimistic pooling, calibrating the subsets to a common scale, and pooling over two neighborhood sizes all fail to improve on the plain mean of Sec. 2.2.

- **The classifier head.** TabPFN’s predicted class probability under the arbitrary context labels is at chance as an anomaly score, and mixing it into the embedding score hurts.
- **Variants of the fine-tune.** Fine-tuning each feature subset separately, fine-tuning through the augmented context, using the pooled isolation estimate as the fine-tune’s trust weights, and contracting each reference sample toward a local center instead of one global center bring no gain.